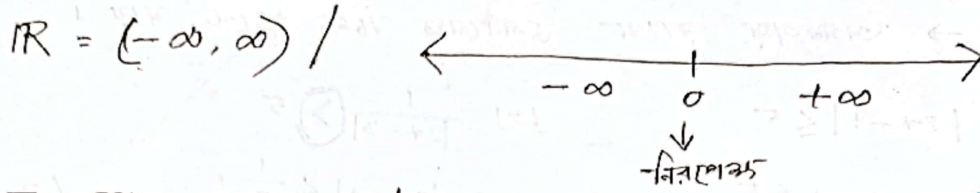


বাস্তব সংখ্যা ও অসীমতা

কি বাস্তব সংখ্যা $\rightarrow$ ক্রটিসহ সংখ্যা দ্বারা তৈরি করা যত বিস্তারিত সংখ্যা সমষ্টি।
কিন্তু অসীমতায় বাস্তব সংখ্যা $\mathbb{R}$ দ্বারা প্রকাশ করা হয়।



কি অসীমতায় বাস্তব সংখ্যা ২ প্রকার।

- $\rightarrow$ মূল সংখ্যা
- $\rightarrow$ অমূল সংখ্যা

কি মূল সংখ্যা $\rightarrow$ যে সকল সংখ্যাকে P/Q আকারে প্রকাশ করা যায়
তাদেরকে মূল সংখ্যা বলে। $P, Q \in \mathbb{Z}, Q \neq 0$ ।

* সকল পূর্ণ সংখ্যা। Ex. 1, 2, 3 $\rightarrow \frac{1}{1}, \frac{2}{1}, \frac{3}{1}$

* অসীম দশমিক সংখ্যা। Ex. 0.5, 0.85 $\rightarrow \frac{5}{10}, \frac{85}{100}$

* অসীম দশমিক সংখ্যা। Ex. $\frac{1}{3} = 0.33333\ldots \rightarrow 0.\dot{3}$
 $0.101101101101\ldots \rightarrow 0.1\dot{0}1$

$\rightarrow$ মূল সংখ্যা Identify করতে পারলেই কোনোটি অমূল সংখ্যা চিহ্নিত করা যায়।
যদি মূল নয় তবেই অমূল।

- $\rightarrow$ ক্রটি $\rightarrow$
- i) open / খোলা $\rightarrow (a, b) \rightarrow$
 - ii) closed / বন্ধ $\rightarrow [a, b] \rightarrow$
 - iii) Open-closed $\rightarrow (a, b] \rightarrow$
 - iv) Closed-open $\rightarrow [a, b) \rightarrow$

কি অসীমতায় বিন্দু $\rightarrow$ i) $|x| < a \rightarrow x \in \mathbb{R}$
 $-a < x < a$

ii) $|x| > a \rightarrow x \in \mathbb{R}$
 $x < -a$ or $x > a$

অসমতার দ্বিঘাত সীমিতকরণ সমাধান :->

$\rightarrow a > b$ প্রক: $(x-a)(x-b) < 0$ প্রক $b < x < a$

$\rightarrow a > b$ প্রক: $(x-a)(x-b) > 0$ প্রক $x < b$ অথবা $x > a$

Ex) $x^2 - 5x + 6 \leq 0$

$\Rightarrow (x-3)(x-2) \leq 0$

$\begin{matrix} + & - \\ - & + \end{matrix}$

$\begin{matrix} + & - & + \\ x-3 \geq 0 & x-2 \leq 0 & \\ \geq 3 & \Rightarrow x \leq 2 & (x-3) \leq 0 & (x-2) \geq 0 \\ & & \Rightarrow x \leq 3 & x \geq 2 \end{matrix}$



$2 \leq x \leq 3$

$x \in [2, 3]$

$\rightarrow x \geq 2$ and $x \leq 3$

$\rightarrow \{x: x \geq 2\} \cap \{x: x \leq 3\}$

উপরে/নিচে ও সর্বোচ্চ/নিম্ন উপরে/নিচে :->

let, $S = \{-10, -5, -3, 0, 3, 8, 13, 20\}$

উপরে/নিচে :-> $[20, \infty)$

সর্বোচ্চ/নিম্ন উপরে/নিচে $\sup(S) = 20$

নিম্ন/উপরে ও সর্বোচ্চ/নিম্ন :->

$S = \{-1, 2, 4, 6\}$

নিম্ন/উপরে = $(-\infty, -1]$

সর্বোচ্চ/নিম্ন উপরে/নিচে $\inf(S) = -1$

$\rightarrow$ অসমতার প্রকার $[] / ()$ ব্যবহৃত হয়।

$\rightarrow$ প্রকারের প্রকার $\{ \}$ ব্যবহৃত হয়।

$\rightarrow$ Infinity প্রক প্রকার Always $()$ হয়।

$$\text{Ex)} \frac{i+i^{-1}}{i-i^{-1}}$$

$$= \frac{i-i}{i+i}$$

$$= 0$$

$$\text{Ex)} \frac{i}{1-\frac{1}{1-i}}$$

$$= \frac{i}{1-\frac{1}{1+i}}$$

$$= \frac{i}{\frac{1+i-1}{1+i}}$$

$$= 1+i$$

$$\text{Ex)} \frac{i-i^{-1}}{i+2i^{-1}}$$

$$= \frac{i-i}{i-2i}$$

$$= \frac{2i}{-i}$$

$$= -2$$

$$\text{Ex)} \frac{i^{-1}-i}{2i^{-1}+i}$$

$$= \frac{-i-i}{-2i+i}$$

$$= \frac{-2i}{-i}$$

$$= 2$$

❖ ସମମૂଳ নির্ণয় :- i (কম্প্লেক্স) সার্বভৌমিক 2 দ্বারা ভাগ করে সার্বভৌমিক উৎপাদক নির্ণয় করা। যদি সত্যের উৎপাদ $-x^2$ হয় i এবং বর্গটির মাঝে $+x^2$ বলে $i^2 = -1$ বোঝানো যায়।

$$\text{Ex)} \sqrt{-7+24i}$$

$$24 \div 2 = 12 = 3 \times 4$$

$$\rightarrow \pm (3+4i)$$

Alt: $-7+24i$
 $= 9+16i^2+2 \cdot 3 \cdot 4i$
 $= (3+4i)^2$

$$\text{Ex)} -8+6i$$

$$\rightarrow \pm (1+3i)$$

$$\text{Ex)} 2i$$

$$\rightarrow \pm (1+i)$$

$$\text{Ex)} -2i$$

$$\rightarrow \pm (1-i)$$

❖ সমমূল নির্ণয় :- নিম্নের দু'সমীকরণে কলকটি পক্ষ সমজাত

$$\text{Ex)} 3i = x$$

$$\Rightarrow x^3 + i^3 = 0$$

$$\Rightarrow (x+i)(x^2-ix-1) = 0$$

$$\therefore x = -i, x = \frac{-i \pm \sqrt{-1+4}}{2}$$

$$= \frac{-i \pm \sqrt{3}}{2}$$

$$\text{Ex)} x = 3i$$

$$\Rightarrow x^3 - i^3 = 0$$

$$\Rightarrow (x-i)(x^2+ix-1) = 0$$

$$\therefore x = i, x = \frac{-i \pm \sqrt{3}}{2}$$

উদাহরণ

$$x = \sqrt{-81} = \sqrt{9i^2}$$

$$\Rightarrow x^4 = (9i^2)$$

$$\Rightarrow x^2 = \pm 9i$$

$$\Rightarrow x = \frac{9}{2}(\pm 2i)$$

$$\Rightarrow x = \frac{9}{2}(1 \pm i)$$

$$\Rightarrow x = \pm \sqrt{\frac{9}{2}}(1 \pm i)$$

$$= \pm \frac{3}{\sqrt{2}}(1 \pm i)$$

$$\text{Ex)} \text{ সমমূলের নির্ণয় :- } \sqrt[4]{-n^2} = \pm \sqrt{\frac{n}{2}}(1 \pm i)$$

$$\text{Ex)} \sqrt[4]{-81} = \sqrt[4]{-9^2} = \pm \sqrt{\frac{9}{2}}(1 \pm i) = \pm \frac{3}{\sqrt{2}}(1 \pm i)$$

❖ জটিল সূত্রের মাধ্যমে সমমূল নির্ণয় :- প্রদত্ত জটিল সংখ্যার অন্তর্গত -১২৪মূল্যের অন্তর্গত নির্ণয়
 করুন এবং তা দিয়ে Main সার্বভৌমিক ভাগ করলে হয় অন্যসংখ্যি যাওয়া যায় (সমীকরণ)

$$\text{Ex)} x = 2+i \text{ হলে } x^4 - 4x^3 + 6x^2 - 4x + 3 = ?$$

$$\Rightarrow x-2 = i$$

$$\Rightarrow x^2 - 4x + 4 = i^2$$

$$\Rightarrow x^2 - 4x + 5 = 0$$

$$\begin{array}{r} x^4 - 4x^3 + 6x^2 - 4x + 3 \\ x^2 - 4x + 5 \end{array} \begin{array}{r} x^4 - 4x^3 + 6x^2 - 4x + 3 \\ x^4 - 4x^3 + 5x^2 \end{array} \begin{array}{r} x^2 - 4x + 3 \\ x^2 - 4x + 5 \end{array} \begin{array}{r} x^2 - 4x + 3 \\ x^2 - 4x + 5 \end{array} \begin{array}{r} x^2 - 4x + 3 \\ x^2 - 4x + 5 \end{array} \begin{array}{r} x^2 - 4x + 3 \\ x^2 - 4x + 5 \end{array}$$

$$\rightarrow i \text{ এর } -\text{সংখ্য } 2 \text{ হলে } x = \text{সুনির্দিষ্ট} - (a^2 + b^2)$$

Q. 1. Find the value of k such that the equation $(k+1)x^2 + 4(k-2)x + 2k = 0$ has equal roots.

1. 3/4
2. 1/4
3. 1/2

$$\begin{aligned} 16(k-2)^2 - 4(k+1)2k &= 0 \\ \Rightarrow 16(k^2 - 4k + 4) - 8k^2 - 8k &= 0 \\ \Rightarrow 16k^2 - 64k + 64 - 8k^2 - 8k &= 0 \\ \Rightarrow 8k^2 - 72k + 64 &= 0 \\ \Rightarrow k^2 - 9k + 8 &= 0 \\ \Rightarrow k^2 - 8k + k + 8 &= 0 \\ \Rightarrow k(k-8) + 1(k-8) &= 0 \\ \Rightarrow (k-8)(k-1) &= 0 \end{aligned}$$

$$i) (k-1)(k-8) > 0$$

$$i) k = 1, 8$$

$$ii) (k-1)(k-8) < 0$$

$$i) 1 < k < 8$$

$$iii) (k-1)(k-8) > 0$$

$$i) k < 1 \text{ or } k > 8$$

Q. 2. Find the value of k such that the equation $3x^2 - kx + 4 = 0$ has equal roots. Find the value of k .

$$k + 12 = -\frac{b}{a}$$

$$k + 12 = \frac{4}{3}$$

$$4k = \frac{k}{3}$$

$$\Rightarrow 12k = k$$

$$k = \frac{4}{3} \Rightarrow k = 8$$

$$k = -\frac{4}{3} \Rightarrow k = -8$$

$$3k = \frac{4}{3}$$

$$\Rightarrow k = \frac{4}{9}$$

$$\therefore k = \pm \frac{2}{3}$$

→ If the equation $3x^2 - kx + 4 = 0$ has equal roots, find the value of k .

Q. 3. Find the value of k such that the equation $x^2 + 2kx + 8 = 0$ has equal roots. Find the value of k .

$$4k = 8$$

$$6 = -a$$

$$\therefore k = 2$$

$$\therefore a = -6$$

$$A) 2, -6$$

$$B) 6, -2$$

$$C) 6, 2$$

$$D) -6, -2$$

Q. 4. Find the value of k such that the equation $x^2 - 5x + k = 0$ has equal roots. Find the value of k .

$$A) 1 \quad B) 2 \quad C) 4 \quad D) -1$$

Lecture - 03 Permutation and Combination

❖ **Permutation** :- જો કોઈ પણ વસ્તુઓ કે અક્ષરોને એકબીજા સાથે ગણવામાં આવે તો તેને Permutation કહેવાય.

❖ **Combination** :- જો કોઈ પણ વસ્તુઓ કે અક્ષરોને એકબીજા સાથે ગણવામાં આવે તો તેને Combination કહેવાય.

→ Permutation અને Combination બંનેમાં જોડવામાં આવેલા વસ્તુઓની ગણના થાય છે.

→ જો કોઈ પણ વસ્તુઓ, અક્ષરો અને કે જે ગણવામાં આવે તેને Permutation કહેવાય.

→ જો કોઈ પણ વસ્તુઓ, અક્ષરો અને કે જે ગણવામાં આવે તેને Combination કહેવાય.

❖ **n** વસ્તુઓમાંથી **r** વસ્તુઓની ગણના થાય છે તો તેને Permutation કહેવાય.

$$\rightarrow nP_r = \frac{n!}{(n-r)!}$$

$$\rightarrow nC_r = \frac{n!}{r!(n-r)!}$$

1! = 1	5! = 120
2! = 2	6! = 720
3! = 6	7! = 5040
4! = 24	8! = 40320

$$\rightarrow n! = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1$$

$$\rightarrow n! = n(n-1)! \quad \left| \quad 5P_2 \rightarrow 5 \times 4 = 20 \right.$$

$$\rightarrow nP_2 = n(n-1)$$

$$\rightarrow nP_n = n! \quad nC_n = 1 \quad \left| \quad 5P_2 \rightarrow \frac{20}{2!} = 10 \right.$$

$$\rightarrow nP_n = nC_n \cdot n! \quad n=1$$

$$\rightarrow nC_x = nC_y \Rightarrow x = y \text{ or } n = x + y$$

$$\rightarrow n > nP_n > nC_n \rightarrow nC_1 + nC_2 + \dots + nC_n = 2^n - 1$$

$$\rightarrow nC_n + nC_{n-1} = {}^{n+1}C_n \quad 5C_3 + 5C_2 = 6C_3$$

$$\text{Ex: } 12C_5 + 12C_6 = 13C_5$$

$$n = 4$$

→ અવકાશીય ગણના :-

$$\text{Permutation} = \frac{n!}{a! b! c! \dots}$$

□ অতিরিক্ত - জিনিস একবার ব্যবহার করে : $\rightarrow {}^n P_n \rightarrow n > r$

Ex | Canada অথ যেক 3 টি করে বা নিচ কয়টি অথ মডি সম থাবে ?
 $\rightarrow {}^6 P_3$

□ নিম্নের কয় : $\rightarrow$ (Optional) ^{Compulsory} $\rightarrow$ $({}^n P_r)$

$\rightarrow$ সুপার নিজস্ব উপায় = (প্রার্থী) সুপার

$\rightarrow$ নিচ এখানে " = (সুপার) নিচ

$\rightarrow$ নিজস্ব এখানে " = (প্রার্থী) নিজস্ব

$\rightarrow$ সেই নিজস্ব মডি এখানে " = (10) নিজস্ব

□ অন্যান্য অন্য : $\rightarrow$

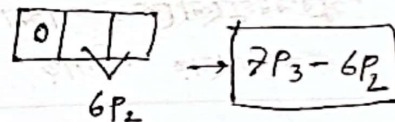
* অতিরিক্ত অন্য একবার ব্যবহার করে .

Ex : 0, 1, 2, 3, 4, 5, 6 অন্য একবার ব্যবহার করে : $\rightarrow$

i) ৩ অন্য কত এক মডি সম থাবে ?

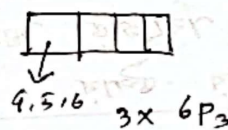
$$\rightarrow {}^7 P_3 - {}^6 P_2 \quad \Bigg/ \quad \begin{array}{l} 6P_1 \times 6P_2 \\ 6 \times 6P_2 \end{array}$$

$$\begin{array}{r} 1 - 6P_1 \\ 2 - \\ 3 - \\ 4 - \\ 5 - \\ 6 - \end{array}$$



ii) 4000 এক কত এক মডি সম থাবে ?

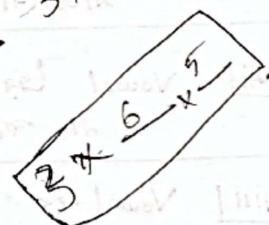
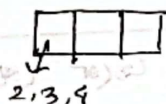
$$\rightarrow 3 \times 6P_3 \quad \begin{array}{r} 4 - 6P_3 \\ 5 - 6P_3 \\ 6 - 6P_3 \end{array}$$



$$3 \times 6 \times 5 \times 4$$

iii) 200 - 500 মডি সম থাবে ?

$$\rightarrow 3 \times 6P_2$$



iv) 4000 এক কত এক মডি সম থাবে এক 5 মডি সম থাবে ?

$$\rightarrow 5 \times 5P_2$$

$$\begin{array}{r} 4 - 5P_2 \\ 5 - \\ 6 - \end{array} \quad \begin{array}{l} 0/5 \\ 0/5 \\ 0/5 \end{array}$$

Ex 1 ଟାକା ଲୋକେ କୁସିନ୍ନା ଗାଆଁର ଲାମ୍ବା 3-ଟି ଏବଂ କୁସିନ୍ନା ଲୋକେ ଟୁଙ୍ଗାଲାର 5-ଟି
ହଳ ଟାକା ଲୋକେ ଟୁଙ୍ଗାଲାର କର ଦିଆଲେ କେତେ ଲାମ୍ବା ଥାନ୍ତେ ?
→ $3 \times 5 = 15$ ଟୁଙ୍ଗାଲାର ।

* ଆନୁସংগিক ବିଶ୍ଳେଷଣ :- ଏହା ଦେଖିବାକୁ ଏହା ସଂସ୍କୃତ ଶବ୍ଦର

D  etc

$$\rightarrow \text{असंगत} \quad \text{१८} \quad = \quad n_1 - n_2$$

→ $2106570 = nC_L (n=2106570)$

$$\rightarrow \text{value} = m c^2 \left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \right)$$

$\text{সমষ্টি } n = n$, $\text{সমষ্টি } m = m$, $\text{অসমষ্টি } n-m = n$
 $\rightarrow \text{সমষ্টি } n / \text{সমষ্টি } m = n-m \subset n-m$

$$F(4) \quad (4) \quad M(6)$$

3

$$\rightarrow 4e_3 \times 5e_0 = 4 \times 5e_3 = 20e_3$$

2 2

$$\rightarrow 4C_2 \times 5C_1 = 30$$

1 3

$$\rightarrow 4C_3 \times 5C_2 = 90$$

0 4

$$\rightarrow 4C_0 \times 5C_3 = \frac{10}{84}$$

$$\rightarrow \text{अविलम्बन} = n-m C_n$$

$\rightarrow 10C_5 = 252$

ଦିଗ୍ଘ ଉପସ୍ଥ (4) (4) ଚାରିପଟ (2)

$$\rightarrow 4 \quad 0 \quad \rightarrow 4C_4 \times 1 = 1$$

$$\rightarrow 3 \quad 1 \quad \rightarrow 4C_3 = 4$$

$$\rightarrow 2 \quad 2 \quad \rightarrow 4C_2 = 6$$

(11)

Ex Degree ଯୁକ୍ତ ଗୁଣିତା 4 କରା ଯାଏ ଯଦି ସିମ୍ପଲ ଗୁଣିତା ଯାହା ଯୁକ୍ତ?

ଦିଗ୍ଘ (3) (4) ଚାରିପଟ (3)

$$\rightarrow 3 \quad 1 \quad \rightarrow 3C_3 = 1$$

$$\rightarrow 2 \quad 2 \quad \rightarrow 3C_2 = 3$$

$$\rightarrow 1 \quad 3 \quad \rightarrow 3C_1 = 3$$

(7)

(Ex) America, Algebra, Abakoni

Total $\rightarrow$ (7)

A $\rightarrow$ 2B $\rightarrow$ 1 Pair

Others $\rightarrow$ 5B

(3)

$$1C_1 \times 5C_1 = 5 \rightarrow 5 \times \frac{3!}{2!} = 15$$

$$6C_3 = 20 \rightarrow 20 \times 3! = 120$$

SMT (25) $\rightarrow$ 25

$$5C_4 \times A \times A \rightarrow 5 \times \frac{3!}{2!}$$

$$6C_3 \rightarrow 20 \times 3!$$

$$(135)$$

(Ex) Examination $\rightarrow$ (4) କ୍ରମାବଳି ସମ୍ପର୍କିତ ଗୁଣିତା ଯାହା ଯୁକ୍ତ?

ତାହା $\rightarrow$ (11) $\rightarrow$ A $\rightarrow$ (2) $\rightarrow$ I $\rightarrow$ (2) $\rightarrow$ N $\rightarrow$ (2) $\rightarrow$ (5) (4)

$$i) 3C_2 \rightarrow 3 \rightarrow 3 \times \frac{4!}{2!2!} = 18$$

$$ii) 3C_1 \times 7C_1 \rightarrow 63 \times \frac{4!}{2!} = 252$$

$$iii) 8C_4 = 70 \rightarrow 70 \times 4! = 1680$$

* Idempotent $\rightarrow$ $A^2 = A$

$$A = \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix}$$

* Nilpotent $\rightarrow$ $A^n = 0$ $\rightarrow$ $A = \begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix} \rightarrow A^2 = 0$

* Involutory $\rightarrow$ $A^2 = I$ $A = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$

* Transpose $\rightarrow$ A^T is the matrix obtained by interchanging the rows and columns of A .

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

* Symmetric $\rightarrow$ $A^T = A$ $A = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} \quad A^T = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$

* Skew Symmetric $\rightarrow$ $A^T = -A$ $\rightarrow$ Main diagonal element is 0.

* Hermitian Matrix $\rightarrow$ $A^H = A$

* Orthogonal $\rightarrow$ $A \cdot A^T = I$

* Adjoint $\rightarrow$ A^H is the conjugate transpose of A .

* Inverse Matrix $\rightarrow$ $A^{-1} = \frac{1}{|A|} \text{Adj}(A)$ $\rightarrow$ $|A| \neq 0$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \text{Adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \rightarrow A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\text{Ex} \quad A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{Find } A^{-1} = ?$$

$$\rightarrow A^{-1} = \frac{1}{4-6} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$\rightarrow$ Matrix A is invertible if $|A| \neq 0$. Elementwise inverse is not possible.

निर्वाचक

→ गणितीयक आदि कलाय उमान १३१० २४।

→ 1. ନିଜ ଘରଟି ନିଜ ଗାଆଁର ଲୋକଙ୍କ ପାଇଁ ଉଦ୍ଦିଷ୍ଟ ହେବା ଉଚିତ ।
2. ଗାଆଁର ଲୋକଙ୍କୁ ଉପଯୋଗୀ କରିବା ଉଚିତ ।
3. ଗାଆଁର ଲୋକଙ୍କୁ ଉପଦେଶ ଦେବା ଉଚିତ ।
4. ଗାଆଁର ଲୋକଙ୍କୁ ଉପକାର କରିବା ଉଚିତ ।
5. ଗାଆଁର ଲୋକଙ୍କୁ ଉପହାସ କରିବା ଉଚିତ ।
6. ଗାଆଁର ଲୋକଙ୍କୁ ଉପହାସ କରିବା ଉଚିତ ।
7. ଗାଆଁର ଲୋକଙ୍କୁ ଉପହାସ କରିବା ଉଚିତ ।
8. ଗାଆଁର ଲୋକଙ୍କୁ ଉପହାସ କରିବା ଉଚିତ ।
9. ଗାଆଁର ଲୋକଙ୍କୁ ଉପହାସ କରିବା ଉଚିତ ।
10. ଗାଆଁର ଲୋକଙ୍କୁ ଉପହାସ କରିବା ଉଚିତ ।

Q. આવક-વિવરણ કે અવકાશ :- $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$

5. $\Delta = \begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} = 9 - 21 = -12$

5. Log 32577K = $(-1)^{2+2} \times \begin{vmatrix} -1 & 3 \\ 7 & 9 \end{vmatrix} = 9 - 21 = -12$

ॐ विनयप्रज्ञा सुदीपिका :- १

→ આર ૦ ફરિ અધિ → ફરિ આગિ થા જાણાએ અહીં ઉપાગત ૦ ફરિ .

$\rightarrow$ ३० " ५५ ५८ ३५७ ६४२ २४ १ ५५५ ५५५

[illegible]

→ " " " " " " " " " "

→ $|A| = |A^T|$ $\left(\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{array} \right) = 0$

Ex) मान नियम $\rightarrow \underline{E_k}$ $\begin{pmatrix} a & a & x \\ c & c & c \\ b & x & b \end{pmatrix} = 0$ जहाँ $x = 9$

$$x = a/b \quad \begin{array}{c} a \\ b \end{array} \quad \left| \begin{array}{c} a \\ b \end{array} \right| \quad \begin{array}{c} a \\ b \end{array}$$

Ex 1

$a+b$	a	b
a	$a+c$	c
b	c	$b+c$

$P = 9$

$a = 1$
 $b = 2$
 $c = 3$

$\rightarrow \begin{vmatrix} 3 & 1 & 2 & 3 & 1 \\ 1 & 4 & 3 & 1 & 4 \\ 2 & 3 & 5 & 2 & 3 \end{vmatrix}$

$$\rightarrow (60 + 6 + 6) - (16 + 27 + 5)$$
$$\Rightarrow 72 - 48 = \boxed{24}$$

☑️ ଉତ୍ତର ମାଧ୍ୟମ ଓ ମାନ ଉପରେ ନିର୍ଭର :-

② $\rightarrow (ax^p + bx^q)^n$ ଉପ ବିକୃତିତ $(n+1)$ ଉତ୍ତର ମାଧ୍ୟମ x^m ବିକୃତିତ
 ଉତ୍ତର ମାଧ୍ୟମ $n = \frac{np - m}{p - q}$

Ex $(x^4 - 2x)^{10}$ ବିକୃତିତ x^{16} ଉତ୍ତର ମାଧ୍ୟମ ଉପରେ ଉତ୍ତର ମାନ ଉପରେ ନିର୍ଭର :-

$n = \frac{20 - 16}{2 - 1} = 4 \rightarrow (4+1)$ ଉତ୍ତର ମାଧ୍ୟମ 5 ଉତ୍ତର ମାଧ୍ୟମ ଉପରେ

ଉତ୍ତର = $10C_4 \cdot (-2)^4$, ମାନ = $10C_4 \cdot (x^4)^6 \cdot (-2x)^4 = 10C_4 \cdot x^{24} \cdot 2^4 \cdot x^4 = 10C_4 \cdot 2^4 \cdot x^{28}$

Ex $(x^4 + \frac{2}{x})^6$ ଉପ ବିକୃତିତ x ଉପରେ ମାନ ଉପରେ ନିର୍ଭର :-

$n = \frac{12 - 0}{2 + 1} = 4 \rightarrow (4+1)$ ଉତ୍ତର / 5 ଉତ୍ତର ମାନ

ମାନ = $6C_4 \cdot 2^4 = 240$ ଉତ୍ତର

☑️ ବିକୃତିତ ଉପରେ ମାନ :-

$\rightarrow (a+x)^n \rightarrow nC_n a^{n-n} \cdot x^n$	$\rightarrow (1+x)^{-2} \rightarrow (-1)^n (n+1) x^n$
$\rightarrow (1+x)^n \rightarrow nC_n x^n$	$\rightarrow (1-x)^{-2} \rightarrow (n+1) x^n (1+x)$
$\rightarrow (1+x)^{-1} \rightarrow (-1)^n x^n$	$\rightarrow (1+x)^{-3} \rightarrow (-1)^n \frac{(n+1)(n+2)}{2} x^n$
$\rightarrow (1-x)^{-1} \rightarrow x^n$	$\rightarrow (1-x)^{-3} \rightarrow \frac{(n+1)(n+2)}{2} x^n$

Ex $\rightarrow (1+x)^{-3}$ ଉପ ବିକୃତିତ x^4 ଉପରେ ମାନ ଉପରେ ନିର୍ଭର :-

$\rightarrow (4+1) \rightarrow \frac{5 \times 6}{2} x^4 = 15x^4$

☑️ ଉପରେ ମାନ :- $\frac{1+x}{1-x}$ ଉପ ବିକୃତିତ x^n ଉପରେ ମାନ ଉପରେ ନିର୍ଭର :- ଉତ୍ତର ମାନ $(1-x)$ ଉପରେ ମାନ $x=1$ ଉପରେ ମାନ x^n ଉପରେ ମାନ ଉପରେ ନିର୍ଭର :-

$\frac{1-x}{1+x}$ ଉପ ବିକୃତିତ x^n ଉପରେ ମାନ ଉପରେ ନିର୍ଭର :-
 $\rightarrow -2$ (ନ ଉପରେ ନିର୍ଭର)
 $\rightarrow 0$ (ନ ବିକୃତିତ ନିର୍ଭର)

Ex $\frac{1+2x+x^2}{1-x}$ ଉପ ବିକୃତିତ x^6 ଉପରେ ମାନ ଉପରେ ନିର୍ଭର :- $1+2+1=4$

$\rightarrow$ ଉତ୍ତର ମାନ $(1-x)^{-2}$ ଉପରେ x^n ଉପରେ ମାନ ଉପରେ ନିର୍ଭର :- $[n+n+1] = 2n+1$

~~Ex~~ $(2+x)^{44}$ ১ ২ ৩ ২২ ৩২ ৪৪ ৪৫ ৪৬ ৪৭ ৪৮ ৪৯ ৫০ $x=9$

$$x = \frac{0.7}{n - (n-1)} \quad 1 = \frac{44 - 21 + 1}{21} \cdot \frac{x}{2} \Rightarrow 21 = 23x \Rightarrow x = \frac{21}{23} = \frac{7}{7}$$

Ex $(1+x)^{24}$ හි x^2 හි සංගුණකය සොයන්න.

$$q = \frac{24 - n + 1}{n} \cdot n \Rightarrow 4n = 25 - n \Rightarrow n = 5$$

5 ଟ 3 6 ଟ

Ex $(1+y)^{21}$ ଓ 11 ଜୁନ 6 12 ଜୁନ ମାତ୍ର ଅବଶ୍ୟକ ଅନ୍ୟତମ ହେଲା $y=9$

$$y = \frac{11}{21 - 11 + 1} = \frac{11}{11} = 1$$

□ Special गिज्ञात x^n को अवस्था :

$$\rightarrow \frac{1}{(1-a)(1-bx)} \text{ ला. विधि से } x^n \text{ का अंश} = \frac{a^{n+1} - b^{n+1}}{a-b}$$

$$\rightarrow \frac{x}{(1-ax)(1-bx)} \quad " \quad " \quad x^n \quad " \quad " \quad = \frac{a^n - b^n}{a-b}$$

Ex1 $\frac{1}{1-5x+x^2}$ ଟାଏଲ୍ ଉନ୍ମୋଚନ କର

$$\rightarrow \frac{(1-3)^{n+1} - 1}{(1-2)(1-3)} \rightarrow \frac{3^{n+1} - 2^{n+1}}{3-2} \rightarrow 3^{n+1} - 2^{n+1}$$

$$\text{Ex} \frac{x}{(1-4x)(1-5x)} \quad \text{or } x^4 \text{ coefficient is } 20$$

$(1-4+)(1-5+)$
 $5^4 - 4^4$
 $5-4$
 $625 - 256 = 369$

□ ଆମିଟର ଏବଂ ଏହା ଆମ ନିର୍ମାଣ : $\rightarrow (1+x)^n$ ଓ ବିକାଶ R U S ଓ

$$R + S = m + 2$$

मध्यस्थता प्रमाणन = $n+2$

→ $A(x_1, y_1)$ ও $B(x_2, y_2)$ বিন্দু দুটির অন্তঃস্থ রেখাংশ x -অক্ষ ও y -অক্ষের উপর

$$x\text{-অক্ষ} = \left| \frac{y_1}{y_2} \right| = |y_1| : |y_2|$$

$$y\text{-অক্ষ} = \left| \frac{x_1}{x_2} \right| = |x_1| : |x_2|$$

Ex $(2, 3)$ ও $(9, 10)$ বিন্দু দুটির অন্তঃস্থ রেখাংশ x -অক্ষের উপর $1:2$ অনুপাতে বিভক্ত করে।
কিভাবে করে? উত্তর: অক্ষাংশের অনুপাত।

$$\begin{array}{ccc} (2, 3) & & (9, 10) \\ & \searrow \quad \nearrow & \\ & 1:2 & \end{array}$$

$$\left(\frac{2+4}{3}, \frac{10+6}{3} \right) = \left(\frac{13}{3}, \frac{16}{3} \right)$$

Ex $A(-3, -2)$ ও $B(6, 4)$ বিন্দু দুটির অন্তঃস্থ রেখাংশ $P(3, 2)$ বিন্দুতে AB অনুপাতে বিভক্ত করে।

$$m_1 : m_2 = -3 - 3 : 3 - 6$$

$$= -6 : -3 = 2 : 1 \text{ Ans.}$$

Ex $(3, -1)$ ও $(8, 9)$ বিন্দু দুটির অন্তঃস্থ রেখাংশ x ও y -অক্ষের উপর AB অনুপাতে বিভক্ত করে।

$$x \rightarrow 1:9$$

$$y \rightarrow 3:8$$

→ (x_1, y_1) ও (x_2, y_2) এর অন্তঃস্থ রেখাংশ $ax + by + c = 0$ রেখা $m_1 : m_2$ অনুপাতে বিভক্ত করে।

$$\frac{m_1}{m_2} = \frac{ax_1 + by_1 + c}{ax_2 + by_2 + c}$$

Ex $(3, -1)$ ও $(8, 9)$ বিন্দু দুটির অন্তঃস্থ রেখাংশ $x - y - 2 = 0$ রেখা $m_1 : m_2$ অনুপাতে বিভক্ত করে।

$$\frac{m_1}{m_2} = \frac{3 + 1 - 2}{8 - 9 - 2} = \frac{2}{-3} \quad | \quad 2:3 \text{ অনুপাতে বিভক্ত করে।}$$

□ ভেক্টর: কীভাবে বিন্দু দুটির মধ্যবিন্দু : → ভেক্টর $A(x_1, y_1)$, $B(x_2, y_2)$ এর মধ্যবিন্দু D হল -

$$x_4 = x_1 + x_2 - x_3$$

$$y_4 = y_1 + y_2 - y_3$$

$$\left| \begin{array}{c} \text{মধ্যবিন্দু} \\ \text{অক্ষাংশ} \end{array} \right| \left| \begin{array}{c} \text{বিন্দু} \\ \text{অক্ষ} \end{array} \right|$$

Ex যদি $A(2, 5)$, $B(5, 9)$, $D(6, 8)$ বিন্দু তিনটি $ABCD$ চতুর্ভুজের মধ্যবিন্দু D এবং C বিন্দু দুটির

$$\begin{array}{l} x_3 = 11 - 2 = 9 \\ y_3 = 17 - 5 = 12 \end{array} \quad (9, 12)$$

→ ଅନୁପାତ :- $n = \frac{abc}{4A}$ ABC ଠ, (ସମତଳ)

Ex ABCD හි $A(3, 0)$, $B(-3, 0)$, $C(0, 4)$ නිසා ලැබෙන්නේ AB මධ්‍යස්ථ රේඛාව AC සමාන්තර වේ.

$$a = \sqrt{3^2 + 4^2} = 5 \quad b = 5 \quad c = 6$$

$$x = \frac{15 - 15}{16} = 0 \quad y = \frac{14}{16} = \frac{7}{8} \rightarrow (0, \frac{7}{8})$$

উদাহরণ : $\rightarrow$ নিচের কোনটি

← असम रक्षा →

□ સામાન્ય સમીકરણ : $ax + by + c = 0$

$\rightarrow ax + by + c = 0$ ରୂପରେ $ax + by = -c$ ରୂପରେ

→ ଏହି ସ୍ତରରେ ଜ୍ଞାନ ଅର୍ଥ ଉପକରଣ ନା ।

→ $a=0$ হলে $\cos 0^\circ = 1$ যা $\sin 0^\circ = 0$ উভয়ের অন্তর্ভুক্ত। $\therefore$ উভয় শ্রেণি নয়।

$\rightarrow b = 0$ " " y " " y " " " / " " " "

[illegible]

Ex1 α ବା 2α ଯାହା ଧନାତ୍ମକ ହେଉ $x^2 + px + (1-p)x + 1 = 0$ ସମୀକରଣର ଉଭୟ ମୂଳ

$C = 1 \Rightarrow 0$
 $\Rightarrow C = 1$

□ ଅବଲମ୍ବନୀୟ ତାଲ : → କେବଳ ଅବଲମ୍ବନୀୟ ୪-ଭିନ୍ନ ବିଭାଗକୁ ନିକଟେ ରଖାଯାଏ

উৎসব রক্ষার tangent এর মান বলা হয়।

→ $m = \tan \theta$ ($\theta = +$ ଉଲ୍ଲସ ବିକଳର ମିଶ୍ରଣ ଆଦି ଓଡ଼ିଆ ଶବ୍ଦ)

$$\rightarrow ax + by + c = 0 \quad \text{Slope } m = -\frac{a}{b} = -\frac{\text{y intercept}}{\text{x intercept}}$$

ସମୀକରଣ (1) ର ସମାଧାନ ନିମ୍ନଲିଖିତ $y = mx + c$ ରୂପରେ ଆପଣଙ୍କ ଗୁରୁତ୍ୱପୂର୍ଣ୍ଣ କାର୍ଯ୍ୟ ସାଧନ

→ (x_1, y_1) ও (x_2, y_2) -বিন্দুসমীপে রেখার ঢাল, $m = \frac{y_1 - y_2}{x_1 - x_2} = \frac{\Delta y}{\Delta x}$

Ex1 $(2, 5)$ ও $(1, 9)$ বিন্দুদ্বয়কে সংযোগকারী রেখাংশের মধ্যবিন্দু (x, y) ।

✓) $x + y - 5 = 0$

B) $2x - 3y + 5 = 0$

c) $x + 2y - 9 = 0$

$$\therefore m = \frac{3-6}{2-1} = -3$$

option can be exercised

मिश्रित : 270° same 2[1]

ফিজিও- টেস্ট এর

Ex) $(2, 3)$ and $(4, 1)$ are two points. Find the equation of the line passing through them.

Ex) $(2, 3)$ and $(4, 1)$ are two points. Find the equation of the line passing through them.

A) $x + y + 5 = 0$ B) $y - 2 = x - 3$ C) $x + 2y - 7 = 0$ D) $x + y - 5 = 0$

Key: i) The slope of the line $m = -\frac{x_1 - x_2}{y_1 - y_2}$

ii) The midpoint of the line $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Ex) Find the equation of the line passing through the points $(2, 3)$ and $(4, 1)$.

$$x = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}$$

$$y = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$$

$$\begin{cases} a_1x + b_1y + c_1 = 0 \\ a_2x + b_2y + c_2 = 0 \end{cases}$$

Ex) Find the equation of the line passing through the points $(2, 3)$ and $(4, 1)$.

Ex) $(2, 3)$ and $(4, 1)$ are two points. Find the equation of the line passing through them.

Key: i) The slope of the line $\rightarrow 3x + 2y - 12 = 0$

ii) The midpoint of the line

$$(x, y) = \left(\frac{36 - 2}{4 + 9}, \frac{3 + 24}{13}\right) = \left(\frac{34}{13}, \frac{27}{13}\right)$$

Ex) Find the equation of the line passing through the points $(2, 3)$ and $(4, 1)$.

$\rightarrow$ The distance between the points $d = \sqrt{a^2 + b^2}$

$\rightarrow$ " " The midpoint of the line (x_1, y_1)

$\rightarrow$ The area of the triangle formed by the line and the axes, $\Delta = \frac{1}{2}ab$

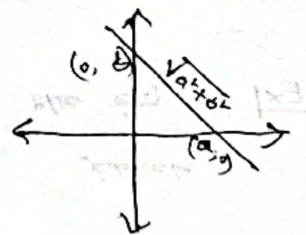
Ex) $3x + 4y - 12 = 0$ is a line. Find the distance between the line and the origin.

The distance between the line and the origin is $d = \frac{|c|}{\sqrt{a^2 + b^2}}$

$\rightarrow$ The distance between the line and the origin $d = \frac{|c|}{\sqrt{a^2 + b^2}} = 3$

$\rightarrow \Delta = \frac{1}{2} \times 12 = 6$

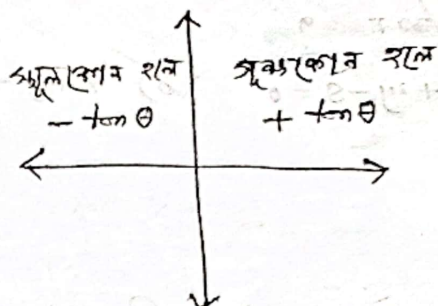
$\rightarrow (x_1, y_1) = (x_2, y_2) = (2, 3)$



$\Delta = \frac{1}{2} \times \text{base} \times \text{height}$
 $ab = c$ then $\Delta = \frac{1}{2}c$

→: ନିମ୍ନ ସମୀକରଣ ଦିଆଯାଇଛି, ତାହାର ସମାଧାନ କର।

$$\tan \theta = \pm \frac{m_1 - m_2}{1 + m_1 m_2}$$



Ex) $3x - y + 7 = 0$ and $x - 3y - 8 = 0$ ଦିଆଯାଇଛି। ତାହାର ସମାଧାନ କର।

- A) $\tan^{-1}(\frac{4}{3})$ B) $\tan^{-1}(-\frac{4}{3})$ C) $\tan^{-1}(\frac{1}{2})$ D) $\tan^{-1}(-\frac{5}{4})$

$$\rightarrow \tan \theta = \frac{3 - \frac{1}{3}}{1 + 1} = \frac{\frac{8}{3}}{2} = \frac{4}{3}$$

$$\begin{cases} m_1 = 3 \\ m_2 = \frac{1}{3} \end{cases}$$

$$\theta = \tan^{-1}(\frac{4}{3})$$

$$\rightarrow \theta = \tan^{-1}(-\frac{4}{3})$$

→ $y = mx + c$ ରେ $y = 6$ ବସାଇବା।

$$\Delta = \frac{1}{2} b^2 \left(\frac{1}{m} - \frac{1}{m_1} \right)$$

Ex1 $2x^2 + 2y^2 + 8x - 12y + 8 = 0$ ରୁ ଡେଇଁ ଲେଖି ନାହିଁ ?
 A) $(-4, 6)$ B) $(-2, 3)$ C) $(4, 2)$ D) $(4, 6)$

Key:
 (-4) ଘାଟ ଘାଟ ଲେଖି ନାହିଁ

Ex1 $3x^2 + ay^2 - ax + 12y + 6 = 0$ ରୁ ଡେଇଁ ଲେଖି ନାହିଁ ?
 A) $(\frac{a}{2}, -6)$ B) $(3, 6)$ C) $(\frac{1}{2}, -2)$ D) $(5, 6)$

Key:
 i) x^2 ଓ y^2 ଲେଖି ନାହିଁ
 ii) -2 ଘାଟ ଘାଟ

Ex1 ଲେଖି ରୁ ଡେଇଁ ଲେଖି ନାହିଁ $(1, 2)$ ଲେଖି ଲେଖି ନାହିଁ ଲେଖି ନାହିଁ $3x + 4y + 1 = 0$ ରୁ ଡେଇଁ ଲେଖି ନାହିଁ ?

$$x = \frac{22}{5} = \frac{3 \cdot 1 + 4 \cdot 2 + 1}{5} = \frac{12}{5}$$

Ex1 $x^2 + y^2 - 2x + 4y - 4 = 0$ ଲେଖି $(1, 1)$ ବିନ୍ଦୁରୁ ଲେଖି ନାହିଁ ଲେଖି ନାହିଁ ଲେଖି ନାହିଁ ?

$$\rightarrow (x, y) = (1, -2)$$

Ex1 $x^2 + y^2 = 16$ ରୁ ଡେଇଁ ଲେଖି ନାହିଁ ଲେଖି ନାହିଁ ଲେଖି ନାହିଁ ?

$$r = \sqrt{0+16} = 4$$

Ex1 $(x+y-2)^2 - 2x + kxy = 0$ ରୁ ଡେଇଁ ଲେଖି ନାହିଁ $k = ?$

$$(k+2)xy = 0 \Rightarrow k = -2$$

Ex1 $x^2 + y^2 + x - y + 2c = 0$ ରୁ ଡେଇଁ ଲେଖି ନାହିଁ $c = ?$

$$(-g, -f) = (-\frac{1}{2}, \frac{1}{2}) \quad 2c = \frac{1}{4} \Rightarrow c = \frac{1}{8}$$

Ex1 ରୁ ଡେଇଁ ଲେଖି ନାହିଁ $(2, 3)$ ଲେଖି ଲେଖି ନାହିଁ $x - 6y + 9 = 0$ ରୁ ଡେଇଁ ଲେଖି ନାହିଁ ?
 A) $x^2 + y^2 + 4x + 6y + 9 = 0$ B) $x^2 + y^2 - 4x - 6y - 4 = 0$ C) $x^2 + y^2 - 4x - 6y + 4 = 0$ D) $x^2 + y^2 - 4x + 6y + 1 = 0$
 Key:
 i) ଲେଖି $(2, 3)$
 ii) $c = 9$

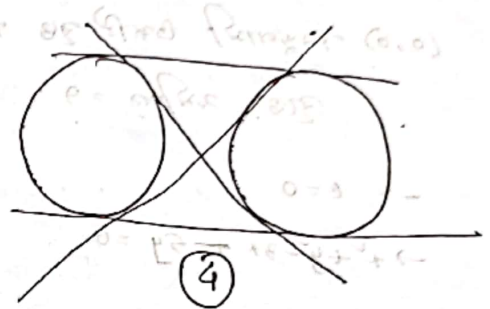
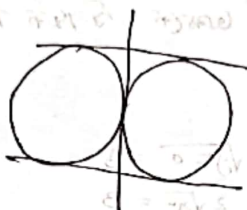
Ex) $y = \sqrt{3}x + 6$ રૂઝાવણી $x^2 + y^2 = 36$ રૂઝાવણી નીચેના રૂઝાવણી

$$\rightarrow \left(\frac{mx}{\sqrt{1+m^2}}, \frac{y}{\sqrt{1+m^2}} \right)$$

Here, $m = \sqrt{3}$
 $c = 6$

$$\rightarrow \left(\frac{6\sqrt{3}}{2}, \frac{6}{2} \right) \rightarrow (3\sqrt{3}, 3)$$

કે રૂઝાવણી નીચેના રૂઝાવણી :



③

④

$$\left(\frac{3\sqrt{3}}{2}, \frac{3}{2} \right) \text{ એ } (0,0)$$

$$\left(\frac{3}{2}, \frac{3\sqrt{3}}{2} \right) \text{ એ } (0,0)$$

$$\frac{3\sqrt{3}}{2} - \frac{3}{2} = \frac{3(\sqrt{3}-1)}{2}$$

$$\frac{3}{2} - \frac{3\sqrt{3}}{2} = \frac{3(1-\sqrt{3})}{2}$$

$$0 = 3 - \sqrt{3} - 3 = 0 \quad 0 = 3 + \sqrt{3} - 3 = 0$$

જો $x^2 + y^2 = 36$ રૂઝાવણી નીચેના રૂઝાવણી

$$m = \frac{y}{x} \quad \frac{1}{m} = \frac{x}{y}$$

$$5x - 2y - 2 = 0$$

Ex] $y^2 = 4x + 8y$ କଣ୍ଟା କଣ୍ଟା କଣ୍ଟା ?

A) (4,4) B) (-4,-4) C) (-4,4) D) (4,-4)

$$\rightarrow y^2 - 8y = 4x$$

$$\Rightarrow y^2 - 2 \cdot 4y + 16 = 4x + 16$$

$$\Rightarrow (y-4)^2 = 4(x+4)$$

$$\rightarrow A(-4, 4)$$

$$2y = 8$$

$$\therefore y = 4$$

$$16 - 32 = 4x$$

$$x = -4$$

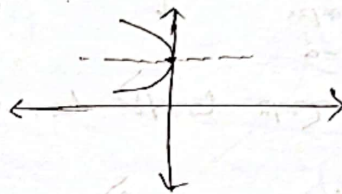
$$(-4, 4) \text{ କଣ୍ଟା}$$

Ex] କଣ୍ଟା କଣ୍ଟା କଣ୍ଟା କଣ୍ଟା ?

* କଣ୍ଟା $\rightarrow$ ଡାହାଣ $\rightarrow +/-$

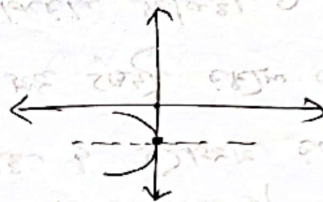
Ex] $(y-2)^2 = -8x$ କଣ୍ଟା କଣ୍ଟା ?

$$\rightarrow (0, 2) \rightarrow x \rightarrow (-)$$



Ex] $x = -(y+1)^2$ କଣ୍ଟା କଣ୍ଟା ?

$$\rightarrow (0, -1) \rightarrow x \rightarrow (-)$$



1st Ex] କଣ୍ଟା କଣ୍ଟା କଣ୍ଟା ?



Ex] $y^2 = -12x$ କଣ୍ଟା କଣ୍ଟା, ଡାହାଣ, ଓ ଡାହାଣ = 3 ଡାହାଣ କଣ୍ଟା କଣ୍ଟା ?

ଓ କଣ୍ଟା କଣ୍ଟା ?

$$a = 3$$

$$(-a, 0) = (-3, 0)$$

$$x = a = 3$$

$$x - a = 0$$

$$x = -a \Rightarrow x = -3 = 0$$

$$\Rightarrow x = -(-3)$$

$$|4a| = 12 \rightarrow x = +(-3) \Rightarrow x + 3 = 0$$

କଣ୍ଟା କଣ୍ଟା

Ex] $(x-2)^2 = -12(y+1)$ କଣ୍ଟା କଣ୍ଟା, ଡାହାଣ, ଓ ଡାହାଣ ?

$$A = (2, -1) \rightarrow S = (x, y) = (0, 3) = (2, 2)$$

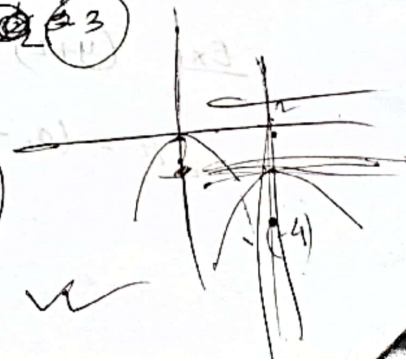
$$a = 3$$

$$\begin{cases} y+1 = 3 \\ y = 2 \end{cases}$$

$$(x-2)^2 = -12(y+1) = (0, -3)$$

$$(x, y) = (2, -4)$$

$$a = 3$$



Ex1 $\frac{x^2}{25} + \frac{y^2}{16} = 1$ এর উৎকেন্দ্রিকতা ও উৎকেন্দ্রের স্থানাঙ্ক কত?

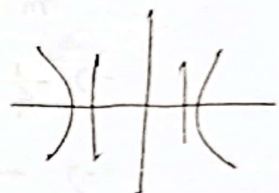
$$e = \sqrt{\frac{25-16}{25}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$$

$$S = (\pm 3, 0)$$

$$\rightarrow \text{উৎকেন্দ্রের স্থানাঙ্ক} = (\pm a, 0)$$

□ **ব্যতিষ্ঠা:** $\rightarrow$ উৎকেন্দ্রিকতার সূত্র: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$e = \sqrt{\frac{a^2+b^2}{a^2}} = \sqrt{\frac{a^2+b^2}{a^2}}$$



$$\rightarrow \text{উৎকেন্দ্র} (\pm ae, 0) \rightarrow \text{উৎকেন্দ্রিকতার সূত্র} = \frac{2b^2}{a^2} \Rightarrow \text{সূত্র: } x = \pm ae$$

$$\rightarrow \text{অসীম উৎকেন্দ্রিকতার সূত্র, } y=0 \text{ এবং উৎকেন্দ্রিকতার সূত্র, } x=0$$

$$\rightarrow \text{দিকবদ্ধতার সূত্র, } x = \pm \frac{a}{e}$$

$$\rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ এর সূত্র } y = mx + c \text{ হলে সূত্র } c = \pm \sqrt{a^2 m^2 - b^2}$$

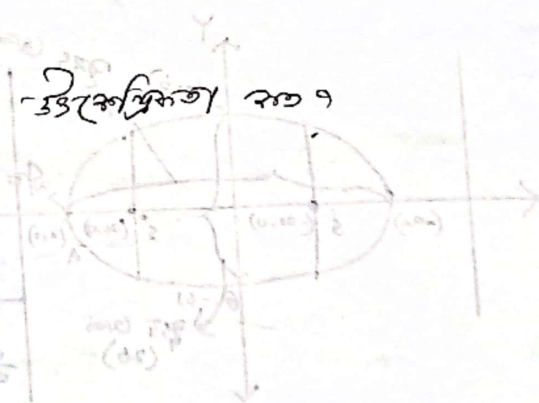
$$\rightarrow xy = a \rightarrow \text{ব্যতিষ্ঠার সূত্র}$$

Ex1 $\frac{x^2}{16} - \frac{y^2}{12} = 1$ এর উৎকেন্দ্রিকতা কত?

$$e = \sqrt{\frac{16+12}{16}} = \frac{\sqrt{28}}{4}$$

Ex1 $y^2 - x^2 = 1$ এর $e = ?$

$$e = \sqrt{\frac{1+1}{1}} = \sqrt{2}$$



~~৭~~ গণিতের মৌলিক ^{1st} function এর Integration এর ক্ষেত্রে একবারে দুই
জানা আছে। গি-রাত का विद्यार्थ जानले एक बार निरंतर उत्तर

$$2 \sin^2 x = 1 - \cos 2x \quad \cancel{2 \cos^2 x} = 1 + \cos 2x$$

$\sin 3x = 3 \sin x - 4 \sin^3 x$

~~Ex 1~~ $\frac{1}{2} \int_2^4 \sin^2 x \, dx = ?$

$$= \frac{1}{2} \int (1 - \cos 2x) dx$$

$$= \frac{1}{2} x - \frac{\sin 2x}{4} + C$$

~~Ex 1~~ $\int \sin^3 t \, dt = ?$

$$= -\frac{3}{4} \cos x + \frac{\cos 3x}{12} + C$$

$$= -\frac{3}{4} \cos x + \frac{\cos 3x}{12} + C$$

~~উদ্ভূত~~ function খোপ বা বিয়োগ আকারে থাকলে $\rightarrow$

$$\int (f_1(x) \pm f_2(x)) dx = \int f_1(x) dx \pm \int f_2(x) dx$$

Ex1 $\int (e^x + \cos x) dx = 9$
 $= e^x + \sin x + C$

Ex1 $\int (x^3 + e^x) dx = 9$ ~~Ex1~~ $\int (\sec^2 x - \sec x \tan x) dx$
 $= \frac{x^4}{4} + e^x + C$ $= \tan x - \sec x + C$

$$= \frac{x^3}{3} + 2x + C$$

$$= \text{format} - \text{secret} + C$$

☐ দুটি function স্থান বা বিমোণ আলাদা এবং তাইলে যদি $\mathbb{R}^*$ দুই আলাদা থাকে:
এবং function দুটি যদি একটি অপসারণ ক্রিয়াকলাপ হয়।

$$\rightarrow \int \underline{e^x} \{ \underline{f(x)} \pm \underline{f'(x)} \} dx = \underline{e^x \cdot f(x)} + C$$

✓ Ex1 $\int e^x (\sin x + \cos x) dx = ?$
 $= e^x \sin x + C$

$$\begin{aligned} \text{Ex 1 } \int e^x (\sin x - \cos x) dx \\ = \int e^x \{ \sin x + (-\cos x) \} dx \\ = -e^x \cos x + C \end{aligned}$$

$$\int e^x \left(\ln x + \frac{1}{x} \right) dx = e^x \ln x + C$$

$$\text{E)} \int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx = \frac{e^x}{x} + C$$

$$\int \frac{x}{(x+1)^2} dx$$

$$= \int e^x \left\{ \frac{1}{x+1} - \frac{1}{(x+1)^2} \right\} dx$$

$$\int e^x (x^2 + 2x) dx = e^x x^2 + C$$

Ex) $\int \frac{1}{(ax+b)\sqrt{cx+d}} dx$ (ଅବକଳ ଫଙ୍କ୍ସନ୍ ଓ $\sqrt{\quad}$ ଓ ଗୁଣାଏ ଫଙ୍କ୍ସନ୍)

ଉଦାହରଣ ଭାବେ 2^{nd} ଉଦାହରଣ ଦେଖ।

Ex) $\int \frac{1}{x+\sqrt{x}} dx$

let $x = z^2$
 $dx = 2z dz$

$= \int \frac{2z}{z^2+2} dz$

$= 2 \int \frac{1}{1+z} dz$

$= 2 \ln(1+\sqrt{x}) + C$

Ex) $\int \frac{dx}{\sqrt{(x-a)(b-x)}} = 2 \sin^{-1} \sqrt{\frac{x-a}{b-a}} + C$

Ex) $\int \frac{dx}{\sqrt{(x-2)(3-x)}} = 2 \sin^{-1} \sqrt{\frac{x-2}{3-2}} = 2 \sin^{-1} \sqrt{x-2} + C$

Ex) ଅବକଳ ଫଙ୍କ୍ସନ୍ ଓ ଗୁଣାଏ ଫଙ୍କ୍ସନ୍ :-

i) $\int \frac{dx}{\sqrt{x^2+a^2}} = \ln(x+\sqrt{x^2+a^2})$

ii) $\int \frac{dx}{\sqrt{x^2-a^2}} = \ln(x-\sqrt{x^2-a^2})$

iii) $\int \sqrt{\frac{a+x}{a-x}} dx = a \sin^{-1} \frac{x}{a} - \sqrt{a^2-x^2}$

$\ln(x+\sqrt{x^2+a^2})$

ସମ୍ଭାବ୍ୟତା (Probability)

କ୍ରମେ Probability ଓ ସମ୍ଭାବ୍ୟତା :->

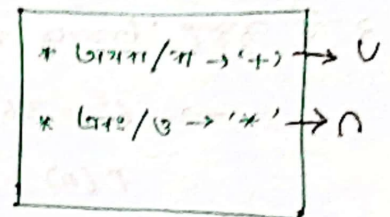
→ Probability ଓ Main formulae ; $P(A) = \frac{\text{ଉପସ୍ଥିତ ଘଟଣା}}{\text{ମୋଟ ଘଟଣା}}$

→ $P(A) + P(A') = 1$

→ ଗୁଣନ ସୂତ୍ର : $P(A \cap B) = P(A) \cdot P(B)$

→ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

→ $0 < P(A) < 1 \rightarrow P(A) \neq 0, P(A) \neq 1$



Ex | $P(A) = \frac{1}{3}, P(B) = \frac{3}{4}, A \cup B$ ଆବଶ୍ୟକ ହେଲେ $P(A \cup B) = ?$

$P(A \cup B) = \frac{1}{3} + \frac{3}{4} - \frac{1}{3} \cdot \frac{3}{4} = \frac{5}{6}$ Ans:

Ex | $P(A \cap B) = \frac{1}{6}, P(A \cup B) = \frac{5}{6}, P(A) = \frac{1}{2}$ ହେଲେ, $P(B) = ?$
 $P(A') = ?$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow \frac{5}{6} = \frac{1}{2} + P(B) - \frac{1}{6}$

$\Rightarrow P(B) = \frac{2}{3}$

$P(A') = 1 - \frac{1}{2} = \frac{1}{2}$

କ୍ରମେ ଖୁଞ୍ଚା ବା ଚୁକ୍କା ସଂଖ୍ୟାଙ୍କ ସମ୍ଭାବ୍ୟତା :->

→ ନମୁନା (କାଠ) : i) ଖୁଞ୍ଚା = $2^n \rightarrow$ ସଂଖ୍ୟାଙ୍କ ନିର୍ଦ୍ଦେଶ / ଖୁଞ୍ଚା ସଂଖ୍ୟା

ii) ଚୁକ୍କା = $6^n \rightarrow n$ " " / ଚୁକ୍କା ସଂଖ୍ୟା

Ex | ଖୁଞ୍ଚା ଓ ଚୁକ୍କା ସଂଖ୍ୟାଙ୍କ ନିର୍ଦ୍ଦେଶ ନେବା ହେଲା । i) ଖୁଞ୍ଚା ସଂଖ୍ୟା ii) ଚୁକ୍କା ସଂଖ୍ୟା
 iii) ଚୁକ୍କା ସଂଖ୍ୟା ଓ ଖୁଞ୍ଚା ସଂଖ୍ୟା ମିଶାଇ ସଂଖ୍ୟାଙ୍କ ସଂଖ୍ୟା

→ ନମୁନା (କାଠ) : $S = \{HH, HT, TH, TT\} \times \{H, T\}$
 $= \{HHH, HHT, HTH, HTT, TTH, THT, TTT\}$

$\rightarrow n(S) = 2^3 = 8$

Ex 1-99 କଣ ଅଛି । ୮ଟି ଖରାଦ ମିଳିଲେ ତା ୫ କଣ ସମ୍ଭବତା ହେବ $P = ?$

$$\rightarrow P = \frac{19}{92}$$

$$* \text{ ସମ୍ଭବତା } = \frac{\text{ପ୍ରାପ୍ତ ହୋଇଥିବା ଖରାଦ}}{\text{ସମସ୍ତ ସମ୍ଭବତା ସଂଖ୍ୟା}}$$

Ex 1-100 କଣ ଅଛି । ୮ଟି ଖରାଦ ମିଳିଲେ ତା ୩୩ କଣ ସମ୍ଭବତା ହେବ $P = ?$

$$P(A) = \frac{33 + 20 - 6}{100} = \frac{47}{100}$$

$$= \frac{33}{100} + \frac{20}{100} - \frac{6}{100}$$

$$= P(A) + P(B) - P(A \cap B)$$

Ex 1-90 କଣ ଅଛି । ୮ଟି ଖରାଦ ମିଳିଲେ ତା ୩୩ କଣ ସମ୍ଭବତା ହେବ $P = ?$

$$\rightarrow \text{ଖରାଦ } P = \frac{9}{90} = \frac{1}{10}$$

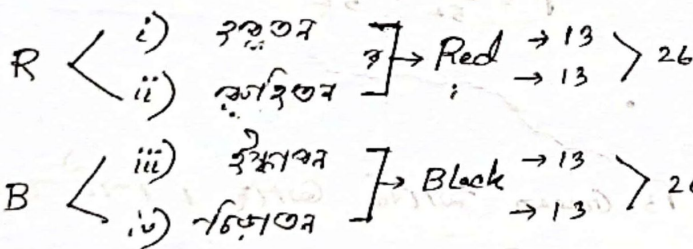
$$\rightarrow \text{ଘରାଦ } P = \frac{4}{90} = \frac{2}{45}$$

$$P = \frac{1}{10} + \frac{2}{45} - \frac{2}{90}$$

Ex 1-520 କଣ ଅଛି । ୮ଟି ଖରାଦ ମିଳିଲେ ତା ୮ କଣ ସମ୍ଭବତା ହେବ $P = ?$

$$\rightarrow P = \frac{8}{520} = \frac{1}{65}$$

Ex ୮ଟି ଖରାଦ ମିଳିଲେ ତା ୫୨ କଣ ସମ୍ଭବତା ହେବ $P = ?$



୫୨ କଣ ସମ୍ଭବତା ହେବ
 କାରଣ: ଖରାଦ, ଘରାଦ, ଖରାଦ, ଘରାଦ
 A, K, Q, J, 10, 9, 8, 7, 6, 5, 4, 3, 2

Ex ୮ଟି ଖରାଦ ମିଳିଲେ ତା ୪ କଣ ସମ୍ଭବତା ହେବ $P = ?$

$$\rightarrow P = \frac{4}{52} = \frac{1}{13}$$

Ex ୮ଟି ଖରାଦ ମିଳିଲେ ତା ୨୪ କଣ ସମ୍ଭବତା ହେବ $P = ?$

$$\rightarrow P = \frac{4}{52} \times \frac{3}{51} = \frac{1}{546}$$

* સુરંગમ્યતા ની કિંમત :-

$$i) P = \frac{100L}{250L} + \frac{150L}{250L} = \frac{1}{2} \quad \Bigg/ \quad P = \frac{10}{25} \times \frac{9}{24} + \frac{25}{25} \times \frac{24}{24} = \frac{1}{2}$$

$$ii) P = \frac{10}{25} \times \frac{15}{24} + \frac{15}{25} \times \frac{10}{24} = 2 \times \frac{10 \times 15}{25 \times 24} = \frac{1}{2} \quad \Bigg/ \quad P = \frac{\text{Num. Colon} \times r}{n(n-1)}$$

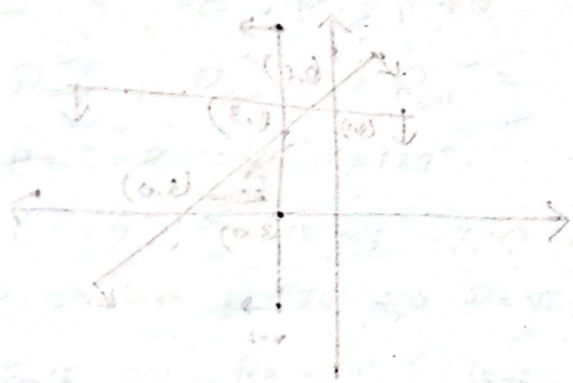
Ex) જાણીએ, 4 Blue, 5 B, 6 R બાદ 1 જોડે જોડે કરી B રજા. $P = 9$

$$\rightarrow P = \frac{5}{15} = \frac{1}{3}$$

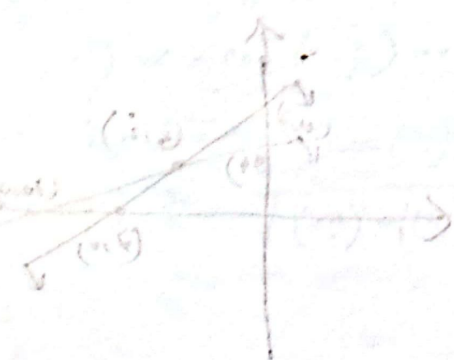
Ex) 10 Blue, 10 R (જુલો 2 જોડે જોડે જોડે કરી. રજા. $P = 9$

$$\rightarrow P = \frac{10}{20} \times \frac{10}{19} + \frac{10}{20} \times \frac{10}{19} = \frac{10}{19}$$

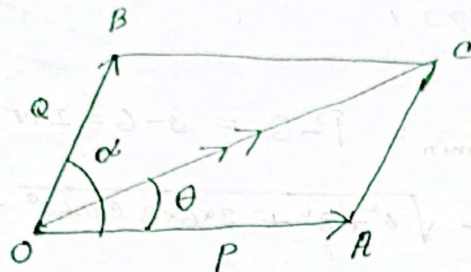
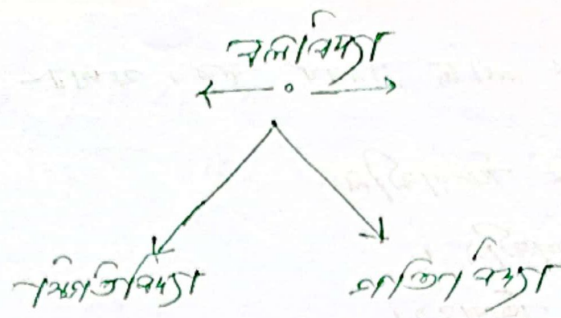
2-0-1-1-5-0	(0,0)
1-1-1-1-5-0	(0,1)
0-1-1-1-5-0	(1,0)
0-0-1-1-5-0	(1,1)



51 = 5	← 0.5
50 = 5	← 0.5
51 = 5	← 0.5



51 = 5	← 0.5
50 = 5	← 0.5
51 = 5	← 0.5



→ ଲମ୍ବ $R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha}$ → ଉତ୍ତରୀୟର ସଂଯୁକ୍ତ ରାଶି ।

→ ଲମ୍ବର ଦିଗ, $\theta = \tan^{-1} \left(\frac{Q \sin \alpha}{P + Q \cos \alpha} \right)$

→ $\alpha = 0^\circ$ ଥିଲେ, $R_{\max} = P + Q$

→ $\alpha = 180^\circ$ ଥିଲେ, $R_{\min} = P - Q$

→ $\alpha = 90^\circ$ ଥିଲେ, $R_{P/90} = \sqrt{P^2 + Q^2}$

→ $R_{\max} + R_{\min} = 2 R_{90}$

→ $P = Q = R$ ଥିଲେ $\alpha = 120^\circ$.

→ $P = 2Q$ ଥିଲେ ଲମ୍ବ ଲମ୍ବ ଉତ୍ତରୀୟର ଉତ୍ତରୀୟ ଉତ୍ତରୀୟ, $\alpha = 120^\circ$

→ $P = Q$ ଥିଲେ $\alpha = 90^\circ$ ଥିଲେ $R = \sqrt{2}P$

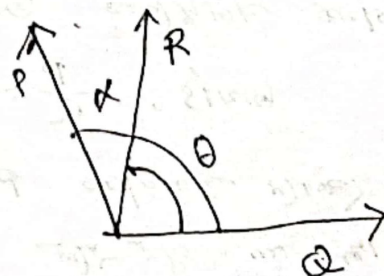
→ $P = Q$ ଥିଲେ, $R = 2P \cos \frac{\alpha}{2}$ ଥିଲେ $\theta = \frac{\alpha}{2}$

→ ଉତ୍ତରୀୟ ଉତ୍ତରୀୟ ଲମ୍ବ R , ଉତ୍ତରୀୟ ଉତ୍ତରୀୟ ଉତ୍ତରୀୟ 90° ଉତ୍ତରୀୟ ଉତ୍ତରୀୟ ଉତ୍ତରୀୟ

i) $\alpha = \cos^{-1} \left(-\frac{Q}{P} \right) \rightarrow \alpha = \cos^{-1} \left(-\frac{1}{2} \right)$

ii) $R = \sqrt{(P \cos \theta)^2 + (P \sin \theta)^2}$

ii) $R = \sqrt{(P \cos \theta)^2 + (P \sin \theta)^2}$



Ex) एकान्तर विद्युत $2p$ ও $3p$ মানে ২টি বল ক্রিয়াশীল। সমাপ্তি ২য় বল সরালে
ও ২য়টির মান ৪ লস্কর হুঁকি সরালে লিঙ্ক এক অপরিবর্তিত থাকে। $p = ?$

$$\frac{2p}{p} = \frac{4p}{p+8}$$

$$\Rightarrow 2p = p+8$$

$$\therefore p = 8 \text{ Ans.}$$

Ex) $3p$ ও $2p$ মানে ২টি বলের লিঙ্ক মান R ; ২য় বলকে বিপরীত করে
লিঙ্ক মানও ২য়র হয়। বলদ্বয়ের মধ্যকার কোণ = ?

$$\cos \alpha = -\frac{3 \cdot 2p}{2 \cdot 2 \cdot 3p} = -\frac{1}{2}$$

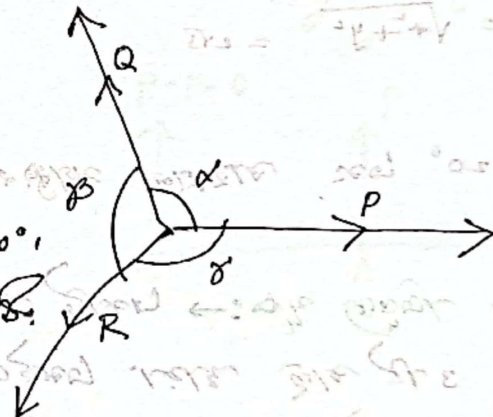
$$\therefore \alpha = 120^\circ$$

উদাহরণ: একান্तर বিদ্যুত ৩টি বলের ভলিউম বল ক্রিয়া করে যদি সমাপ্তি
হুঁকি করে তবে তাদের প্রত্যেকটির মান অন্যর দুটির অন্তর্গত একান্तर মারিত্বমানসিদ্ধ।

$$\rightarrow \frac{P}{\sin \beta} = \frac{Q}{\sin \gamma} = \frac{R}{\sin \alpha}$$

* বলদ্বয় মানে হলে তাদের মধ্যকার কোণ 120° ।

* ৩টি বল সমাপ্তি হলে একান্तर ২টি
লিঙ্ক অন্তর্গত মানে হয়।



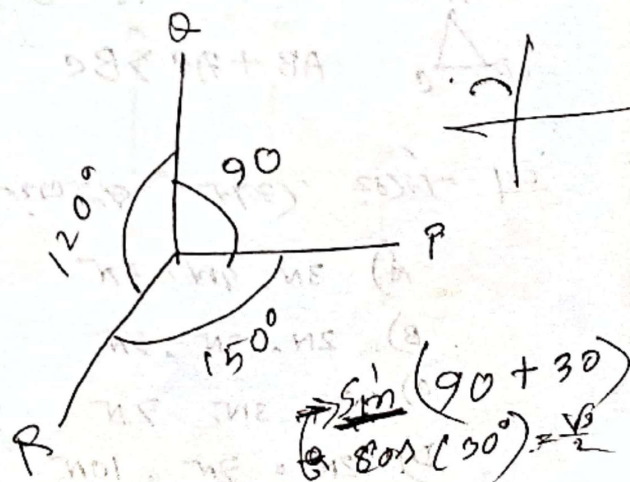
Ex) একান্तर বিদ্যুত ক্রিয়াকারী বল দুটি হুঁকি করলে মানে। তাদের ২য় ও ২য়টির
মধ্য কোণ 90° হলে ২য় ও ২য়টির মধ্য কোণ 120° হলে বলদ্বয়ের অনুপাত = ?

$$\frac{P}{\sin 120^\circ} = \frac{Q}{\sin 150^\circ} = \frac{R}{\sin 90^\circ}$$

$$\Rightarrow \frac{P}{\frac{\sqrt{3}}{2}} = \frac{Q}{\frac{1}{2}} = \frac{R}{1}$$

$$\Rightarrow P:Q:R = \frac{\sqrt{3}}{2} : \frac{1}{2} : 1$$

$$= \sqrt{3} : 1 : 2$$



Ex. 32cm කුණාටු 2 ක් තිබුණ 15kg & 9kg ප්‍රමාණයෙන්, 2 ක් තබාගන්නා තරම තුළින් ගැලේ, ගැලේ බරින් ලැබෙන බර කීය?

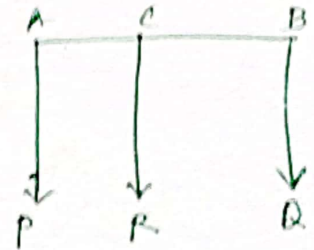
→ 32cm කුණාටු 2 ක් තිබුණ 15kg & 9kg ප්‍රමාණයෙන්, 2 ක් තබාගන්නා තරම තුළින් ගැලේ, ගැලේ බරින් ලැබෙන බර කීය?

$$\rightarrow R = P + Q$$

$$\rightarrow P \cdot P \text{ ලෙස බරින් දුන්න} = Q \cdot Q \text{ ලෙස බරින් දුන්න}$$

$$\Rightarrow P \cdot x = Q \cdot y$$

$$\Rightarrow \frac{P}{BC} = \frac{Q}{AC} = \frac{R}{AB}$$

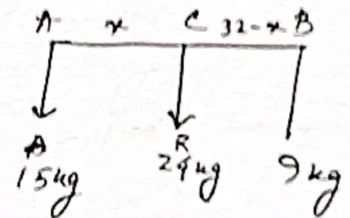


Ex. 32cm කුණාටු 2 ක් තිබුණ 15kg & 9kg ප්‍රමාණයෙන්, 2 ක් තබාගන්නා තරම තුළින් ගැලේ, ගැලේ බරින් ලැබෙන බර කීය?

$$\text{බරින් } R = 15 + 9 = 24 \text{ kg}$$

$$15x = 9(32-x)$$

$$\Rightarrow x = 12 \text{ cm}$$

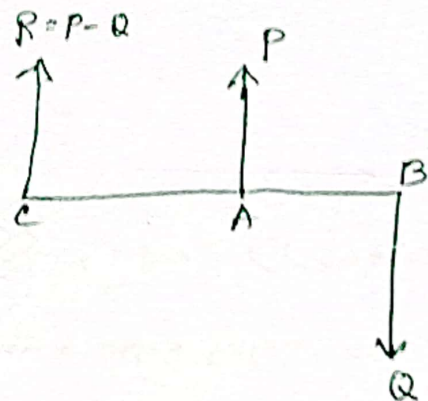


Ex. තිබුණ තරම, ලැබෙන බර කීය?

$$\rightarrow R = P - Q$$

$$\rightarrow \frac{P}{BC} = \frac{Q}{AC} = \frac{R}{AB}$$

$$\rightarrow \frac{P}{Q \leftarrow \frac{d}{QR} \rightarrow R} = \frac{Q}{P \leftarrow \frac{d}{PQ} \rightarrow R} = \frac{R}{P \leftarrow \frac{d}{PQ} \rightarrow Q}$$



Ex. Previous Math :-

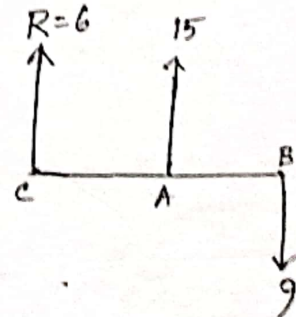
$$\frac{15}{BC} = \frac{9}{AC} = \frac{R}{AB} = \frac{6}{32}$$

$$\Rightarrow \frac{15}{BC} = \frac{6}{32}$$

$$\frac{9}{AC} = \frac{6}{32}$$

$$\therefore BC = 80 \text{ cm}$$

$$\Rightarrow AC = 48 \text{ cm}$$



$$\text{Ex} \quad \lim_{x \rightarrow \infty} \frac{x^2 - 4}{x + 2}$$

$$= \lim_{x \rightarrow \infty} \frac{1 - \frac{4}{x^2}}{\frac{1}{x} + \frac{2}{x^2}}$$

$$= \frac{1}{0} = \infty$$

$$\text{Ex} \quad \lim_{x \rightarrow \infty} \frac{x + 7}{2x^2 + 5}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x} + \frac{7}{x^2}}{2 + \frac{5}{x^2}}$$

$$= \frac{0}{2} = 0$$

$$\text{Ex} \quad \lim_{x \rightarrow \infty} \frac{1 - 2x + 3x^2}{5 - x + 2x^2}$$

$$= \frac{3}{2} \text{ Ans}$$

$$\lim_{x \rightarrow \infty} \frac{Aa^x - Ba^{-x}}{Ca^x + Da^{-x}} = \frac{A}{C}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{a+bx} - \sqrt{c+dx}}{x} = \frac{b-d}{2}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{a+x} - \sqrt{a-x}}{x} = \frac{1}{\sqrt{a}}$$

Ex) $f\left(\frac{1}{x}\right) = \frac{1}{1+x}$ and $f(2) = ?$

Ex) $f(x+2) = x+1$ and $f(x) = ?$

$\boxed{\frac{1}{x} = 2}$
 $\Rightarrow x = \frac{1}{2}$
 $f\left(\frac{1}{x}\right) = \frac{1}{1+\frac{1}{x}}$
 $\Rightarrow f(2) = \frac{1}{\frac{2}{2}} = \frac{2}{3}$

$\Rightarrow f(x-2+2) = x-2+1$ | $y = x+2$
 $\therefore f(x) = x-1$ | $\Rightarrow x = y-2$
 $= x-2$

Ex) $f(x) - f(x^2-1) = x+2$ and $f(x) = ?$

$\Rightarrow f(x) = \sqrt{x+1} + 2$ | $y = x^2-1$
 $\Rightarrow x = \sqrt{y+1}$
 $\Rightarrow x = \sqrt{x+1}$

Ex) $f(x^3-3) = x-2$ and $f(x)$ and $f(8) = ?$

$f(x) = \sqrt[3]{x+3} - 2$ | $y = x^3-3$
 $\Rightarrow x = \sqrt[3]{y+3}$
 $f(8) = \sqrt[3]{8+3} - 2$
 $= \sqrt[3]{11} - 2 = 2-2 = 0$

□ Inverse function exists: $\rightarrow$

$y = f(x)$
 $\downarrow$ Range
 $\downarrow$ Domain
 $\rightarrow x = f^{-1}(y)$

$f(x) \leftrightarrow f^{-1}(x)$
 $D = R$
 $R = D$

$\rightarrow f(x) = ax+b$ and $f^{-1}(x) = \frac{x-b}{a}$

$\rightarrow f(x) = ax^2+b$ and $f^{-1}(x) = \sqrt{\frac{x-b}{a}}$

$\rightarrow f(x) = ax^m+b$ and $f^{-1}(x) = \sqrt[m]{\frac{x-b}{a}}$

$\rightarrow f(x) = \frac{ax+b}{cx+d}$ and $f^{-1}(x) = \frac{-dx+b}{cx-a}$ [Note: $a \leftrightarrow d$, $b \leftrightarrow c$, $c \leftrightarrow -a$, $d \leftrightarrow -b$]

Ex) $f(x) = 3x-6$

$f^{-1}(x) = \frac{x+6}{3}$

Ex) $f(x) = 3x^2-1$

$f^{-1}(x) = \sqrt{\frac{x+1}{3}}$

Ex) $f(x) = \frac{2}{2x+3}$

$f^{-1}(x) = \frac{3x+2}{2x}$

□ One-One Function: $\rightarrow$ Domain has unique image and Co-Domain has unique pre-image. If both are true, it is a 1-1 function.

$\rightarrow$ Range = Co-Domain $\rightarrow$ Inverse function exists.

$\rightarrow$ Function is 1-1 if and only if it is 1-1.

$\rightarrow$ Absolute function is not 1-1.

$\rightarrow$ Trigonometric function is not 1-1.

$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

Ex1 $f(x) = x^2 - 5x + 3$ Find Domain and Range = ?

Domain = $(-\infty, \infty) / \mathbb{R}$

Range = $[3 - \frac{25}{4}, \infty) = [-\frac{13}{4}, \infty)$

$* \sqrt{x^2} = |x|$

Ex1 $f(x) = 6x - 2x^2 + 1$ Find Domain and Range = ?

Domain = $\mathbb{R} / (-\infty, \infty)$

Range = $(-\infty, 1 + \frac{36}{8}] = (-\infty, \frac{44}{8}] = (-\infty, \frac{11}{2}]$

✓ Ex1 $f(x) = x^2$ Find D and R = ?

Ex1 $f(x) = \sqrt{x^2 + 3}$ Find D and R = ?

D = $\mathbb{R} / (-\infty, \infty)$

R = $[0, \infty)$

$y = x^2$
 $\Rightarrow x = \sqrt{y}$
 $\Rightarrow f^{-1}(y) = \sqrt{y}$
 $\therefore D = [0, \infty)$

D = $\mathbb{R}$

R = $[3, \infty)$

$y = \sqrt{x^2 + 3}$
 $x = \sqrt{y^2 - 3}$

Ex1 Root function Example $f(x) = \sqrt{x^2 - 9}$ Find Domain and Range of Real value function. Root function is defined for $x \geq 0$ and $x \leq 0$ and $x = 0$.

Ex1 $f(x) = \sqrt{3x+4}$ Find D and R = ?

Ex1 $f(x) = \sqrt{x^2 - 9}$ Find D and R = ?

D: $3x+4 \geq 0$
 $\Rightarrow x \geq -\frac{4}{3}$

$\rightarrow D = [-\frac{4}{3}, \infty)$

$\rightarrow R = [0, \infty)$

$x^2 - 9 \geq 0$

$\Rightarrow x^2 \geq 9$

$x \leq -3$ or $x \geq 3$

D = $(-\infty, -3] \cup [3, \infty)$

R = $[0, \infty)$

* $f(x) = \sqrt{9 - x^2}$ Find Domain and Range = ?

$9 - x^2 \geq 0$
 $\Rightarrow -x^2 \geq -9$
 $\Rightarrow x^2 \leq 9$

$-3 \leq x \leq 3$

D = $[-3, 3]$

R = $[0, \sqrt{\text{constant}}]$

= $[0, \sqrt{9}]$

= $[0, 3]$

~~$f(x) = \sqrt{25 - x^2}$~~

Ex) $f(x) = \frac{3x+6}{2x-1}$

$D = \mathbb{R} - \left\{ \frac{1}{2} \right\}$

$R = \mathbb{R} - \left\{ \frac{3}{2} \right\}$

Ex) $f(x) = \frac{x+3}{x+2}$

$D = \mathbb{R} - \{-2\}$

$R = \mathbb{R} - \{1\}$

→ লব্ধি ব্লক বসে শূন্য ব্লক বসানো অসম্ভব: →

$f(x) = \frac{0x+c}{ax+b}$

$D = \mathbb{R} - \left\{ -\frac{b}{a} \right\}$

$f'(x) = \frac{-bx+c}{ax}$

Range of $f(x)$ is $\mathbb{R} - \{0\}$

→ লব্ধি ব্লক বসে শূন্য ব্লক বসানো অসম্ভব: →

$f(x) = \frac{ax^2+b}{cx+d}$

$cx+d \neq 0$

$\Rightarrow x = -\frac{d}{c}$

$D = \mathbb{R} - \left\{ -\frac{d}{c} \right\}$

$\lim_{x \rightarrow -\frac{d}{c}} \frac{ax^2+b}{cx+d}$

$= \lim_{x \rightarrow -\frac{d}{c}} \frac{2ax+\cancel{b}}{c} = \frac{2a(-\frac{d}{c})+\cancel{b}}{c}$

$= \frac{-2ad+\cancel{b}}{c} = \frac{-2ad}{c}$

Ex) $f(x) = \frac{2x^2+8}{x+1}$

$x+1 \neq 0$

$\Rightarrow x \neq -1$

$D = \mathbb{R} - \{-1\}$

$\lim_{x \rightarrow -1} \frac{2x^2+8}{x+1}$

$= \lim_{x \rightarrow -1} \frac{4x+\cancel{8}}{1}$

$= -4$

$R = \mathbb{R} - \{-4\}$

Exponential function $\Rightarrow f(x) = e^{ax+b}$

$D = \mathbb{R}$

$y = e^{ax+b}$

$\Rightarrow \ln y = ax+b$

$\Rightarrow ax = \ln y - b$

$\Rightarrow x = \frac{\ln y - b}{a}$

$\Rightarrow f^{-1}(x) = \frac{\ln x - b}{a}$

$R = (0, \infty)$

← →

- ସାମ୍ବାଦିକ ଲେଖାଟି ସୂକ୍ଷ୍ମ ଗ୍ରାହ୍ୟ ।
- ୧୫-୨୦ ଭାଗର ସାମ୍ବାଦିକର ସାମ୍ବାଦିକତା ଲେଖା, ଗ୍ରନ୍ଥ, କଥାକଥାକ୍ତି ଇତ୍ୟାଦି ଗ୍ରନ୍ଥାବଳୀରୁ ଅନ୍ୟତମ ସାମ୍ବାଦିକତାରେ ସଂକ୍ଷିପ୍ତ ଭାବେ ଲେଖାଯାଇଥାଏ ।
- ସାମ୍ବାଦିକ ଲେଖାଟି ସାମ୍ବାଦିକ ଲେଖକଙ୍କ ଦ୍ଵାରା ଲେଖାଯାଇଥାଏ ।
- ସାମ୍ବାଦିକ ଲେଖାଟି ସାମ୍ବାଦିକ ଲେଖକଙ୍କ ଦ୍ଵାରା ଲେଖାଯାଇଥାଏ ।

ଉଦାହରଣ :- ସାମ୍ବାଦିକ ଲେଖାଟି

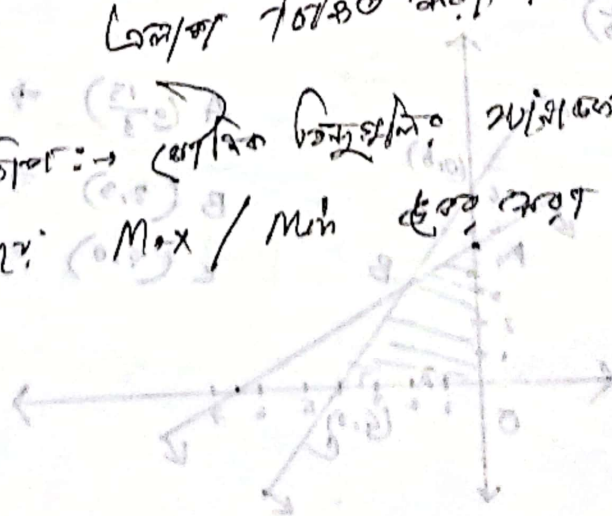
୧. ସାମ୍ବାଦିକ ଲେଖାଟି :- ସାମ୍ବାଦିକ ଲେଖକଙ୍କ ଦ୍ଵାରା ଲେଖାଯାଇଥାଏ ।

୨. ସାମ୍ବାଦିକ ଲେଖାଟି :- ସାମ୍ବାଦିକ ଲେଖକଙ୍କ ଦ୍ଵାରା ଲେଖାଯାଇଥାଏ ।

୩. ସାମ୍ବାଦିକ ଲେଖାଟି :- ସାମ୍ବାଦିକ ଲେଖକଙ୍କ ଦ୍ଵାରା ଲେଖାଯାଇଥାଏ ।

୪. ସାମ୍ବାଦିକ ଲେଖାଟି :- ସାମ୍ବାଦିକ ଲେଖକଙ୍କ ଦ୍ଵାରା ଲେଖାଯାଇଥାଏ ।

୫. ସାମ୍ବାଦିକ ଲେଖାଟି :- ସାମ୍ବାଦିକ ଲେଖକଙ୍କ ଦ୍ଵାରା ଲେଖାଯାଇଥାଏ ।



Ex) Find $Z_{\max} = 3x + 4y$ subject to constraints

S/t :-

$$\begin{aligned} x + y &\leq 7 \\ 2x + 5y &\leq 20 \\ x &\geq 0 \\ y &\geq 0 \end{aligned}$$

$x + y = 7$ --- (i)

$\Rightarrow \frac{x}{7} + \frac{y}{7} = 1$

$(7, 0) \quad (0, 7)$

$2x + 5y = 20$ --- (ii)

$\Rightarrow \frac{x}{10} + \frac{y}{4} = 1$

$\therefore (10, 0) \quad (0, 4)$

$x = 0$ --- (iii)

$y = 0$ --- (iv)

$O(0, 0) \rightarrow Z = 0$

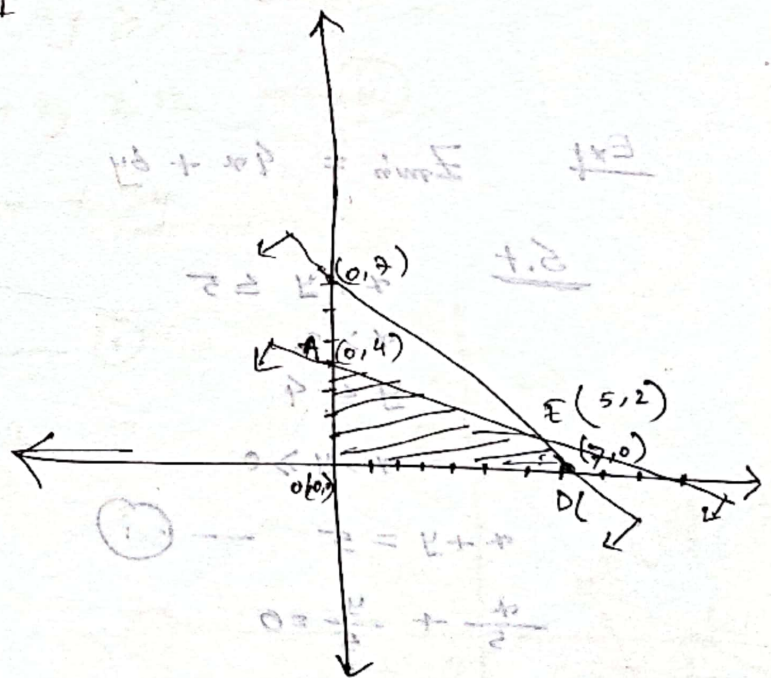
$A(0, 4) \rightarrow Z = 16$

$E(5, 2) \rightarrow Z = 23$ max

$B(7, 0) \rightarrow Z = 21$

$D(10, 0) \rightarrow Z = 30$

$C(0, 7) \rightarrow Z = 28$



(i) $x = 0$

(ii) $y = 0$

Ex 1

ଆବାହ	ପ୍ରୋଡକ୍ଟ	ଆୟତ	ବିଲୋପାଦିତ
A	1	3	2 ଟଙ୍କା
B	3	2	3 ଟଙ୍କା
ଦୈନିକ ମୁରବୀ ଅବଶ୍ୟକ	9	12	-

Let, A ଟଙ୍କା x ଟଙ୍କା, ଆୟତ y ଟଙ୍କା
 ଟଙ୍କା ଆୟତ (ଅବଶ୍ୟକ) ମୁରବୀ ହେଉ ।

By Condition,

$$\text{ପ୍ରୋଡକ୍ଟ : } x + 3y \geq 9 \quad \text{--- (i)}$$

$$\text{ଆୟତ : } 3x + 2y \geq 12 \quad \text{--- (ii)}$$

$$\text{ଆୟତ ସର୍ବନିମ୍ନ : } Z = 2x + 3y$$

$$x + 3y = 9 \quad \text{--- (1)}$$

$$\Rightarrow \frac{x}{9} + \frac{y}{3} = 1$$

$$(9, 0) \text{ ଓ } (0, 3)$$

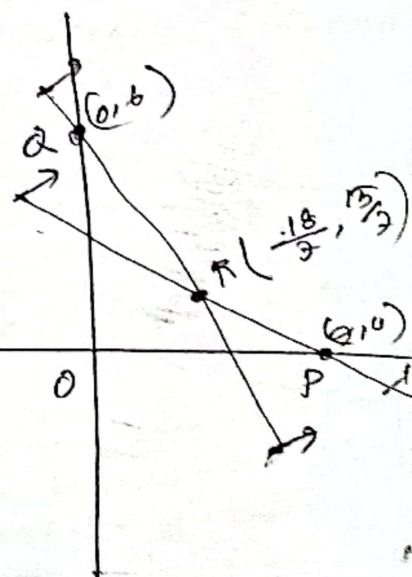
$$3x + 2y = 12 \quad \text{--- (ii)}$$

$$\Rightarrow \frac{x}{4} + \frac{y}{6} = 1$$

$$\therefore (4, 0) \text{ ଓ } (0, 6)$$

$$\text{--- (0, 4) --- Z =}$$

$$R\left(\frac{18}{5}, \frac{16}{5}\right) = 11\frac{4}{5} \text{ ଟଙ୍କା}$$



□ त्रिकोणमितीय सूत्र व्युत्पत्ति :-

$$\begin{cases} * \sin(A+B) = \sin A \cos B + \cos A \sin B & \text{--- (i)} \\ * \sin(A-B) = \sin A \cos B - \cos A \sin B & \text{--- (ii)} \end{cases}$$

$$\begin{cases} * \cos(A+B) = \cos A \cos B - \sin A \sin B & \text{--- (iii)} \\ * \cos(A-B) = \cos A \cos B + \sin A \sin B & \text{--- (iv)} \end{cases}$$

(iii) + (iv) and (iii) - (iv)

$$\begin{aligned} \text{(i) + (ii)} \quad \text{(i) - (ii)} \\ * \sin(A+B) + \sin(A-B) &= 2 \sin A \cos B \quad \text{(5)} \\ * \sin(A+B) - \sin(A-B) &= 2 \cos A \sin B \quad \text{(6)} \end{aligned}$$

From $\rightarrow$ (iii) $A=B$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\Rightarrow \cos 2A = 1 - 2 \sin^2 A$$

$$\Rightarrow 2 \sin^2 A = 1 - \cos 2A$$

$$\Rightarrow \cos 2A = 2 \cos^2 A - 1$$

$$\Rightarrow 2 \cos^2 A = 1 + \cos 2A$$

From (i) $\Rightarrow$ If $B=A$

$$\rightarrow \sin 2A = 2 \sin A \cos A$$

Let, $A+B=C$
 $A-B=D$

$$A = \frac{C+D}{2}, B = \frac{C-D}{2}$$

From (5) (6) (7) (8) $\Rightarrow$

$$* \sin C + \sin D = 2 \sin \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right)$$

$$* \sin C - \sin D = 2 \cos \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right)$$

$$* \cos C + \cos D = 2 \cos \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right)$$

$$* \cos C - \cos D = 2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{D-C}{2} \right)$$

$$\begin{aligned} * \cos 2\theta &= \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \\ * \tan 2\theta &= \frac{2 \tan \theta}{1 - \tan^2 \theta} \end{aligned}$$

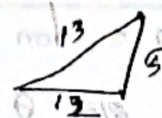
$$\begin{aligned} * \sin 3A &= \\ * \cos 3A &= \end{aligned}$$

$$* \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$* \tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Ex) If $\cos A = \frac{12}{13}$ then $\tan A = ?$

$$\rightarrow \tan A = \frac{5}{12}$$



$\rightarrow$ Condition of question $\pm$ given option is correct Ans

☐ ନିମ୍ନ ଉଲ୍ଲେଖିତ Direct ଫର୍ମୁଲା : Awla - jhawa

Ex | $\frac{\cot \theta + \sin \theta}{\cot \theta - \sin \theta}$

= $\frac{1 + \tan \theta}{1 - \tan \theta}$

= $\frac{\tan 45^\circ + \tan \theta}{1 - \tan 45^\circ \tan \theta}$

= $\tan (45^\circ + \theta)$

= $\tan 53$

Ex | $\frac{\sin 15^\circ + \cos 15^\circ}{\cos 15^\circ - \sin 15^\circ}$

= $\frac{\tan 45^\circ + \tan 15^\circ}{1 - \tan 45^\circ \tan 15^\circ}$

= $\tan (60^\circ) = \sqrt{3}$

Ex | $\frac{\sin 75^\circ + \sin 15^\circ}{\sin 75^\circ - \sin 15^\circ}$

= $\frac{\cos 15^\circ + \sin 15^\circ}{\cos 15^\circ - \sin 15^\circ}$

= $\tan (45^\circ + 15^\circ) = \sqrt{3}$

Ex | $\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} \div \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$

→ ନିମ୍ନୋକ୍ତ ସମ୍ବନ୍ଧକୁ ଯୋଗ କରି ଦିଆଯାଇଛି

→ $\frac{\cos A + \sin A}{\cos A - \sin A} = \tan (45^\circ + A)$ and

$\frac{\cos A - \sin A}{\cos A + \sin A} = \tan (45^\circ - A)$

→ $A + B = 90^\circ$ ଥିଲେ $\tan A - \tan B = 2 \tan (A - B)$

→ $\sin A + \cos A = \sqrt{2} \cos (A - 45^\circ) = \sin (A + 45^\circ)$

Ex | $\sin 65^\circ + \cos 65^\circ = \sqrt{2} \cos (65^\circ - 45^\circ) = \sqrt{2} \cos 20^\circ$

Ex | ABC ତ୍ରିଭୁଜରେ $\cos^2 A + \cos^2 B + \cos^2 C = 9$ ଥିଲେ

$A = \frac{\pi}{2}$ and $B + C = \frac{\pi}{2}$

$\cos^2 \frac{\pi}{2} + \cos^2 B + \cos^2 (\frac{\pi}{2} - B) = 0 + \cos^2 B + \sin^2 B = 1$

→ $\sin A + \cos A = \sin B + \cos B$ ଥିଲେ $A + B = \frac{\pi}{2}$

Ex 1.2.2 :-

$$i) \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$$

$$ii) \tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$$

$$iii) \sec^{-1}x + \csc^{-1}x = \frac{\pi}{2}$$

$$\text{Ex 1} \quad \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}$$

$$= \tan^{-1} \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{6}}$$

$$= \tan^{-1} \left(\frac{\frac{5}{6}}{\frac{5}{6}} \right)$$

$$= \frac{\pi}{4} \text{ or } 45^\circ$$

$$\text{Ex 1} \quad \sin \left\{ 2 \left(\sin^{-1}x + \cos^{-1}x \right) \right\} = 9$$

$$= \sin \left(2 \times \frac{\pi}{2} \right) = 0$$

$$\text{Ex 1} \quad \sec^{-1}\frac{1}{8} + \csc^{-1}\frac{1}{8} = \frac{\pi}{2}$$

$$\rightarrow \tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 = \pi$$

$$\text{Ex 1} \quad \tan^{-1}x + \tan^{-1}2 + \tan^{-1}3 = \pi$$

$$\Rightarrow \tan^{-1}x + \tan^{-1}\frac{5}{1-b} = \pi$$

$$\Rightarrow \tan^{-1}x = \tan^{-1}1 = \pi$$

$$\Rightarrow \tan^{-1}\frac{x-1}{1+x} = \tan^{-1}1$$

$$\therefore x = 1$$

$$\text{Ex 1} \quad \sec(\tan^{-1}2) + \csc(\cot^{-1}3)$$

$$= 1 + \left\{ \tan(\tan^{-1}2) \right\}^2 + 1 + \left\{ \cot(\cot^{-1}3) \right\}^2$$

$$= 1 + 4 + 1 + 9$$

$$= 15$$

$$\text{Ex 1} \quad \sin^{-1}x + \sin^{-1}y = \frac{\pi}{2} \quad \text{2nd} \quad x^2 + y^2 = 9$$

$$\therefore \sin^{-1}x = \frac{\pi}{4} \quad \sin^{-1}y = \frac{\pi}{4}$$

$$\therefore x = \frac{1}{\sqrt{2}} \quad y = \frac{1}{\sqrt{2}}$$

$$\therefore x^2 + y^2 = 1$$

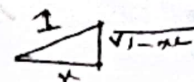
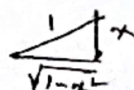
$$\text{Ex 1} \quad \cos \tan^{-1} \cot \sin^{-1}x$$

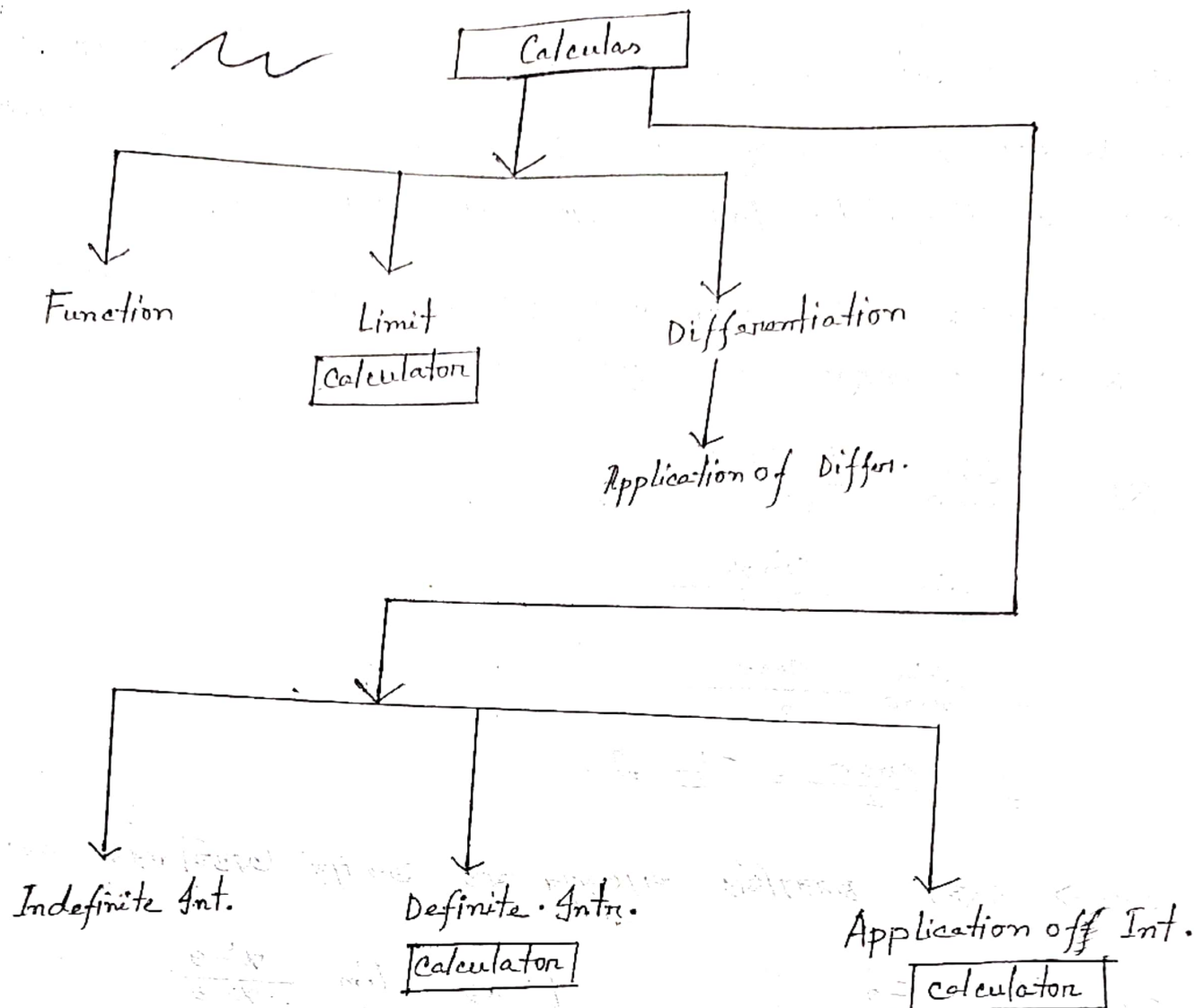
$$= \cos \tan^{-1} \cot \cot^{-1} \frac{\sqrt{1-x^2}}{x}$$

$$= \cos \tan^{-1} \frac{\sqrt{1-x^2}}{x}$$

$$= \cos \cot^{-1}x$$

$$= x$$





Limit

Q. L'Hôpital's Rule: $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1}$

$$= \frac{\text{numerator derivative}}{\text{denominator derivative}} \times \left(\frac{0}{0} \right)$$

Q. $\lim_{x \rightarrow a} \frac{x^{\frac{9}{2}} - a^{\frac{9}{2}}}{\sqrt{x} - \sqrt{a}}$

$$= \lim_{x \rightarrow a} \frac{x^{\frac{9}{2}} - a^{\frac{9}{2}}}{x^{\frac{1}{2}} - a^{\frac{1}{2}}} = \frac{\frac{9}{2}}{\frac{1}{2}} \cdot a^{\frac{9}{2} - \frac{1}{2}} = 9a^4 \text{ Ans.}$$

$\lim_{x \rightarrow 0} \text{Größt}$ $\lim_{x \rightarrow \infty} (1+ax)^{\frac{1}{x}}$ $\text{Ergebnis} \rightarrow e^{ax}$

$\lim_{x \rightarrow 0} \frac{1}{x}$ ଅସୀମ $\lim_{x \rightarrow \infty} \frac{1}{x}$ $\lim_{x \rightarrow 0} x$ $\lim_{x \rightarrow \infty} x$

Q $\lim_{x \rightarrow 0} (1 + 5x)^{\frac{3}{x}} = ?$

$$= e^{15}$$

Q/ $\lim_{x \rightarrow \infty} (1+2x)^{\frac{3x+5}{x}} = ?$

$$= \lim_{x \rightarrow \infty} (1 + 2x)^{3 + \frac{5}{x}}$$

$$= e^{2 \times 5}$$

$$= e^{10} \text{ A.}$$

☐ $\lim_{x \rightarrow \infty} (\text{শিফন্যানিটিক সারি}) : \rightarrow$ হয় উল্লম্বাচি-থাকছে, সমষ্টি

উল্লেখ্য যে নিচের অংশটি খসড়া লিখিত মান ২৫৬

ତା'ର ଅବସ୍ଥା ଦୃଷ୍ଟିକାରେ !

→ limit ବୋଲେ, Infinity (∞) ଅର୍ଥ ଅସୀମ ।

$$\underline{Q)} \quad \lim_{x \rightarrow \alpha} \frac{x^2 - 9}{x + 7}$$

$$= \frac{1}{0} = \alpha \quad Q.$$

$$= \frac{1}{0} = \infty \text{ Q.}$$

Q1 $\lim_{x \rightarrow \infty} \frac{x+2}{2x^2+5}$

$$x \rightarrow \infty \quad \frac{2x^2 + 5}{x^2 + 1}$$

$$= \frac{0}{2} = 0 \quad \text{Q.E.D.}$$

$$\frac{0}{2} = 0 \quad \text{Q.E.D.}$$

এই অক্যালকুলেটর সফটওয়্যার :-> অক্যালকুলেটর সফটওয়্যার থাকলে
আরও সহজে অক্যালকুলেটর সফটওয়্যার ব্যবহার করা যায় $\frac{\sin \theta}{\theta}$ বা $\frac{\tan \theta}{\theta}$ বা
 $\frac{\cos \theta}{\theta}$ ইত্যাদি সফটওয়্যার সফটওয়্যার সফটওয়্যার সফটওয়্যার। অথবা L'Hospital's
Rule সফটওয়্যার সফটওয়্যার।

$$\begin{aligned} \text{Q/ } \lim_{x \rightarrow 0} \frac{\sin 5x}{7x} &\rightarrow \lim_{x \rightarrow 0} \frac{\sin mx}{nx} = \frac{m}{n} \\ &= \lim_{x \rightarrow 0} \frac{5}{7} \times \frac{\sin 5x}{7x} \\ &= \frac{5}{7} \text{ Ans.} \end{aligned}$$

1. Since the calculator is not available, we will use the limit and infinity.

→ ଜ୍ୟୋତ୍ସ୍ନା ସମୟର ଆକାଶଟି ଡେରିଆ Ca/C ଚାଲିଯାଏ । ତା'ର ବାତର
ହେଉଥିବା ସମୟରେ ଆକାଶଟି ଡେରିଆ ଚାଲିଯାଏ । ତା'ର ବାତର

1. ଆମର ସମସ୍ତ ସମ୍ପଦ

→ গিতিক/সমিতিক Function এর লক্ষণ Calculator এক অবস্থায় Rad mode এ সম্পন্ন হয়।

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$u \frac{dv}{dx} + v \frac{du}{dx}$$

Q1 $y = x^3 \cdot e^x$ Find $\frac{dy}{dx} = ?$

$$\therefore \frac{dy}{dx} = x^3 \cdot e^x + e^x \cdot 3x^2$$

$$= e^x \cdot x^2 (x + 3) \text{ Q.}$$

Q1 $\frac{d}{dx} \{ e^x \ln x \}$

$$= \{ \frac{e^x}{x} + e^x \ln(x) \}$$

$$= 5e^x \left(\frac{1}{x} + \ln x \right) \text{ Q.}$$

Q1 $y = e^x \cdot \cos x$ Find $\frac{dy}{dx} = ?$

$$\therefore \frac{dy}{dx} = e^x \cdot -\sin x + \cos x \cdot e^x$$

$$= e^x (\cos x - \sin x) \text{ Q.}$$

Q1 $y = \frac{\sin x + 1}{1 + \cos x}$

$$\therefore y = \frac{1 - \cos x}{1 + \cos x}$$

$$\Rightarrow y = 1 - \cos x$$

$$\therefore \frac{dy}{dx} = \sin x \text{ Q.}$$

Q1 $\frac{d}{dx} \left\{ \frac{2}{3} \sin x \cdot \ln(x) \right\}$

$$= \frac{2 \sin x}{3x} + \frac{2 \cos x}{3} \ln x$$

$$= \frac{2}{3} \left\{ \frac{\sin x}{x} + \cos x \cdot \ln x \right\} \text{ Q.}$$

Q1 $\frac{d}{dx} \left(\frac{1+x+x^2}{x} \right) = ?$

$$= \frac{d}{dx} \left(\frac{1}{x} + 1 + x \right)$$

$$= \left(-\frac{1}{x^2} + 0 + 1 \right)$$

$$= 1 - \frac{1}{x^2} \text{ Q.}$$

Q1 $\frac{d}{dx} \left(\frac{\sin x}{x} \right) = ?$

$$= \frac{x \cos x - \sin x}{x^2}$$

Q1 $\frac{d}{dx} \left(\frac{1 + \sin x}{1 + \cos x} \right) = ?$

Q1 $\frac{d}{dx} \left(\frac{\ln x}{\sin x} \right) = ?$

$$= \frac{\sin x \cdot \frac{1}{x} - \ln x \cos x}{\sin^2 x}$$

$$= \frac{\frac{\sin x}{x} - \ln x \cos x}{\sin^2 x} \text{ Q.}$$

$$= \frac{(1 + \cos x) \cos x + (1 + \sin x) \sin x}{(1 + \cos x)^2}$$

$$= \frac{\cos x + \cos^2 x + \sin x + \sin^2 x}{(1 + \cos x)^2}$$

$$= \frac{1 + \sin x + \cos x}{(1 + \cos x)^2}$$

✓ Q. x ଓ y ଗୁଡ଼ିକର ଉପରାସ୍ତା ଅନୁସାରେ ଉପରାସ୍ତା କରାଯାଉ,

$$\frac{dy}{dx} = \frac{y - \text{ଗୁଡ଼ିକର ଉପରାସ୍ତା}}{x - \text{ଗୁଡ଼ିକର ଉପରାସ୍ତା}}$$

$$\frac{dx}{dy} = \frac{x - \text{ଗୁଡ଼ିକର ଉପରାସ୍ତା}}{y - \text{ଗୁଡ଼ିକର ଉପରାସ୍ତା}}$$

$$\boxed{\frac{dy}{dx}} = \frac{y - f^n}{x - f^n}$$

✓ Q. $x = \cos t$ ଗୁଡ଼ିକର $y = 1 - \sin t$ ଥିଲେ, $\frac{dy}{dx} = ?$

$$\therefore \frac{dy}{dx} = \frac{0 - \cos t}{-\sin t} = \cot t \quad \text{Q.}$$

✓ Q. $x = \cos \theta$, $y = \cos \theta + \sin \theta$ ଥିଲେ $\frac{dy}{dx} = ?$

$$\therefore \frac{dy}{dx} = \frac{-\sin \theta + \cos \theta}{-\sin \theta} = 1 - \cot \theta \quad \text{Q.}$$

✓ Q. $x = a(\theta - \sin \theta)$ ଗୁଡ଼ିକର $y = a(1 - \cos \theta)$ ଥିଲେ, $\frac{dy}{dx} = ?$

$$\therefore \frac{dy}{dx} = \frac{\sin \theta}{1 - \cos \theta} = \frac{\cancel{2} \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\cancel{2} \sin^2 \frac{\theta}{2}} = \cot \frac{\theta}{2}$$

$$\sin 1\theta = 2 \sin \theta \cos \theta$$

✓ Q. $x = \sin \theta$ ଗୁଡ଼ିକର $y = \cos \theta$ ଥିଲେ, $\frac{dy}{dx} = ?$

$$\therefore \frac{dy}{dx} = \frac{-\sin \theta}{\cos \theta} = -\tan \theta \quad \text{Q.}$$

✓ Q. $x = e^t \cos t$ ଗୁଡ଼ିକର $y = e^t \sin t$ ଥିଲେ, $\frac{dy}{dx} = ?$

$$\therefore \frac{dy}{dx} = \frac{e^t \cos t + \sin t e^t}{-e^t \sin t + e^t \cos t} = \frac{\sin t + \cos t}{\cos t - \sin t} \quad \text{Q.}$$

Q1) $y = e^{x^x}$ find $\frac{dy}{dx} = ?$

$$\Rightarrow \ln y = \ln e^{x^x}$$

$$\Rightarrow \ln y = x^x \ln e$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = x^x \left\{ \frac{x}{x} \frac{dx}{dx} + \ln x \cdot \frac{dx}{dx} \right\}$$

$$\Rightarrow \frac{dy}{dx} = e^{x^x} \cdot x^x (1 + \ln x)$$

Q2) $x^y = e^{x-y}$ find $\frac{dy}{dx}$

$$\Rightarrow y \ln x = x - y$$

$$\Rightarrow y \ln x - x + y = 0$$

$$\therefore \frac{dy}{dx} = \frac{-\left(\frac{y}{x} - 1\right)}{\ln x + 1} = \frac{x-y}{x(1+\log x)}$$

Q3) $y = \log_x (a+x)$ find $\frac{dy}{dx} = ?$

$$\Rightarrow x^y = a+x$$

$$\Rightarrow \frac{d}{dx} (x^y) = \frac{d}{dx} (a+x)$$

$$\Rightarrow x^y \frac{d}{dx} (y \ln x) = 1$$

$$\Rightarrow \frac{y}{x} + \frac{dy}{dx} \ln x = x^{-y}$$

$$\therefore \frac{dy}{dx} = \frac{x^{-y} - \frac{y}{x}}{\ln x}$$

$$\frac{dy}{dx} = \frac{x^{-y} - \frac{y}{x}}{\ln x}$$

$$\log_{10} 1000 = 3$$

$$10^3 = 1000$$

$$\log_2 5 = x$$

$$\log_2 8 = 3$$

$$y \ln x = \ln(a+x)$$

$$y \cdot \frac{1}{x} + \ln x \frac{dy}{dx} = \frac{1}{a+x}$$

$$\ln x \frac{dy}{dx} = \frac{1}{a+x} - \frac{y}{x}$$

$$\boxed{\text{Q1}} \quad 2 \tan^{-1} x = \tan^{-1} \frac{2x}{1-x^2} = \cos^{-1} \frac{1-x^2}{1+x^2} = \sin^{-1} \frac{2x}{1+x^2}$$

$$\text{Q1} \quad y = \tan^{-1} \frac{2x}{1-x^2}$$

$$\Rightarrow y = 2 \tan^{-1} x$$

$$\Rightarrow y_1 = 2 \cdot \frac{1}{1+x^2} = \frac{2}{1+x^2}$$

$$\text{Q1} \quad y = \cos^{-1} \frac{1-x}{1+x} = \cos^{-1} \frac{1-(\sqrt{x})^2}{1+(\sqrt{x})^2} = 2 \tan^{-1} \sqrt{x}$$

$$\therefore y_1 = \frac{1}{1+(\sqrt{x})^2} \cdot \frac{1}{\sqrt{x}} = \frac{1}{\sqrt{x}(1+x)}$$

$$\text{Q2} \quad \begin{aligned} \text{i) } \sin 3x &= 3 \sin x - 4 \sin^3 x \\ \text{ii) } \cos 3x &= 4 \cos^3 x - 3 \cos x \end{aligned} \quad \sin 3x =$$

$$\text{Q1} \quad y = \cos^{-1} (4x^3 - 3x) \quad \text{3rd} \quad \frac{dy}{dx} = ?$$

$$\Rightarrow y = 3 \cos^{-1} x$$

$$\therefore \frac{dy}{dx} = -\frac{3}{\sqrt{1-x^2}}$$

$\frac{d}{dx} (4x^3 - 3x) = 12x^2 - 3$
 $\frac{d}{dx} \cos^{-1} (4x^3 - 3x) = \frac{-1}{\sqrt{1-(4x^3-3x)^2}} (12x^2 - 3)$
 $\frac{d}{dx} \cos^{-1} (4x^3 - 3x) = \frac{-3(4x^2 - 1)}{\sqrt{1-(4x^3-3x)^2}}$
 $\frac{d}{dx} \cos^{-1} (4x^3 - 3x) = \frac{-3(4x^2 - 1)}{\sqrt{1-16x^6+24x^3-9x^2}}$
 $\frac{d}{dx} \cos^{-1} (4x^3 - 3x) = \frac{-3(4x^2 - 1)}{\sqrt{1-16x^6+24x^3-9x^2}}$
 $\frac{d}{dx} \cos^{-1} (4x^3 - 3x) = \frac{-3(4x^2 - 1)}{\sqrt{1-16x^6+24x^3-9x^2}}$
 $\frac{d}{dx} \cos^{-1} (4x^3 - 3x) = \frac{-3(4x^2 - 1)}{\sqrt{1-16x^6+24x^3-9x^2}}$

→ Helping chart →

Term	Do	Term	Do
$1+x^2$	$x = \tan \theta$	$\frac{2x}{1-x^2}$	$x = \tan \theta$
$1-x^2$	$x = \sin \theta$	$\frac{1-x^2}{1+x^2}$	$x = \tan \theta$
$\sqrt{\frac{1-x}{1+x}}$	$x = \cos \theta$		

$$\frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2} \quad 2 \tan^{-1} x =$$

Q1 $x^2 + y^2 - 2x - 3 = 0$ चक्रीयता/र र्हा आसत विस्तार मरम कइएन
+ कइत आसत/उतर आसत = ?

$$\frac{dy}{dx} = -\frac{2x-2}{2y} = \frac{-(x-1)}{y}$$

x 6/23/23 3/21/23/24 3/21/24, $\frac{dy}{dx} = 0$
 $\Rightarrow \frac{-(x-1)}{y} = 0$

$$x=1$$

ଅନ୍ତରାଳ ଗଣନା ପଦ୍ଧତି, $y = \pm 2$

$\therefore$ বিকল্প অংগভুক্ত = $(1, \pm 2)$ ও

④ ସ୍ୱଚ୍ଛତା ଓ ନୀତିଗତ ନିର୍ଦ୍ଦେଶ :-

2nd step : $\frac{dy}{dx}$ এর মান বের করা।

$$\frac{dy}{dx} = 0$$

$$g = \theta$$

2nd step : $\frac{dy}{dx} = 0$ ରେ x ର ମୂଲ୍ୟ ନିର୍ଣ୍ଣୟ କର ।

End Step: ଏ ଓଡ଼ିଆ ଶାସ୍ତ୍ର ଓଡ଼ିଆ ସ୍ୱରାଜ୍ୟ ଓଡ଼ିଆ ଏ ଓଡ଼ିଆ
ଏ ଓଡ଼ିଆ ଶାସ୍ତ୍ର ଓଡ଼ିଆ ସ୍ୱରାଜ୍ୟ ଓଡ଼ିଆ ଏ ଓଡ଼ିଆ

Q1. * ତିନି ସହ ସଠ କଲି, * $(12-2x)^{-1}$ ର ସଂଯୁକ୍ତ ଓ ସଂଯୁକ୍ତ

$$y = x(12 - 2x)^2$$

$$\Rightarrow y = 4x(x^2 - 12x + 36)$$

$$\Rightarrow y = 4x(x^2 - 12x + 36)$$

$$\therefore \frac{dy}{dx} = 4(x^2 - 12x) + 36 + 9x(2x - 12) \cdot \left(\frac{d}{dx}(2x-12) \right) = 4x^2 - 48x + 144 + 8x^2 - 72x$$

$$= 12x^2 - 96x + 144$$

$$= 12(x^2 - 8x + 12)$$

$$= 12(x-6)(x-2)$$

$$f' = 0$$

Q. a^x ର n ଥର ଉତ୍ତର ଅଟେ :

$$y = a^x \text{ ହେଲେ, } y_n = a^x \cdot (\log_e a)^n$$

$$y = \frac{1}{3x+2} \quad y_4 = (-1)^4 \cdot 4! \cdot (3x+2)^{-5}$$

Q. $y = 3^x$ ହେଲେ, $\frac{d^3 y}{dx^3} = ?$

$$y_3 = 3^x (\log_e 3)^3 \text{ ଉ.}$$

Q. $\frac{1}{x}$ ର n ଥର ଉତ୍ତର ଅଟେ $\Rightarrow$

$$y = \frac{1}{x} \text{ ହେଲେ, } y_n = (-1)^n \cdot n! \cdot x^{-(n+1)}, \frac{d(x)}{dx}$$

Q. $y = \frac{1}{ax+b}$ ହେଲେ, $y_3 = ?$ $\frac{3!}{(ax+b)^{-1}}$

$$y_3 = (-1)^3 \cdot (3!) (ax+b)^{-4} \cdot a^3 \text{ ଉ.} = (-1)^3 (ax+b)^{-4} \cdot a^3$$

$$= (-1)^3 \cdot 3! (ax+b)^{-4}$$

$\rightarrow$ ଟାଣ୍ଡେମ୍ ଟାଣ୍ଡେମ୍ ଅକ୍ସାଇଡ୍ ଅକ୍ସାଇଡ୍ $\Rightarrow$

Q. $y = c + (1+x)$ ରେକ୍ଟାଙ୍ଗୁଲାର୍ ଅକ୍ସାଇଡ୍ ଟାଣ୍ଡେମ୍ ଅକ୍ସାଇଡ୍ ଟାଣ୍ଡେମ୍ ଅକ୍ସାଇଡ୍ ଅକ୍ସାଇଡ୍ 30° ଟାଣ୍ଡେମ୍ ଅକ୍ସାଇଡ୍ ଅକ୍ସାଇଡ୍, $c = ?$

$$\frac{dy}{dx} = c(1+x) \text{ ; } (0,0) \text{ ଟାଣ୍ଡେମ୍ } \frac{dy}{dx} = c$$

$$\therefore \frac{dy}{dx} = \tan 30^\circ = \left(\frac{1}{\sqrt{3}} \right)$$

$$(-1)^n \cdot n! \cdot x^{-(n+1)}$$